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Overcoming the uncertainty challenges in detecting building changes from remote sensing images

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ABSTRACT

Detecting building changes with multi-temporal remote sensing (RS) imagery at a very high resolution can help us understand urbanization and human activities, making informed decisions in urban planning, resource allocation, and infrastructure development. However, existing methods for building change detection (BCD) generally overlook critical uncertainty phenomena presented in RS imagery. Specifically, these uncertainties arise from two main sources: First, current building change detection datasets are designed primarily to detect changes in buildings, while changes in other land-cover classes are often classified as an unchanged background. Because of the manual labeling process, background elements that resemble buildings, such as roads and bridges, are at significant risk of being misclassified as building changes, introducing aleatoric uncertainty at the data level. Second, changes in parts of buildings that affect appearance, texture, or style without altering their semantic meaning, known as pseudo-changes, along with the imbalance between changed and unchanged samples, together lead to epistemic uncertainty at the model level. To address these challenges, we present an Uncertainty-Aware BCD (UA-BCD) framework. In detail, we employ a Siamese pyramid vision transformer to extract multi-level features from bi-temporal images, which are then decoded via a general decoder to obtain a coarse change map with inherent uncertainty. Subsequently, we introduce the aleatoric uncertainty estimation module to estimate the aleatoric uncertainty and embed it into the feature space. Then, a knowledge-guided feature enhancement module is developed to leverage the knowledge encoded in the coarse map to enhance the multi-level features and generate a refined change map. Finally, we propose an epistemic uncertainty estimator that takes the bi-temporal images and the refined change map as input and outputs an estimate of epistemic uncertainty. This estimation is supervised by the entropy calculated from the refined map, ensuring that the UA-BCD framework can produce a change map with lower epistemic uncertainty. To comprehensively validate the efficacy of the UA-BCD framework, we adopt a dual-perspective verification approach. Extensive experiments on five public building change datasets demonstrate the significant advantages of the proposed method over current state-of-the-art methods. Additionally, an application in Dongxihu District, Wuhan, China, confirms the outstanding performance of the proposed method in large-scale BCD. The source code of the project is available at https://github.com/Henryjiepanli/UA-BCD.

1. Introduction

With the increasing availability of very high-resolution (VHR) remote sensing (RS) imagery (Huang et al., 2024), the precise detection of surface changes within designated areas has become feasible, opening up a vast array of applications across diverse domains. Buildings, as fundamental man-made structures in urban environments, play a key role in urban development and societal advancement (Li et al., 2023). Accurate detection and monitoring of building changes with multi-temporal RS images are essential for effective urban planning (Chen et al., 2024), comprehensive environmental monitoring (Zhang et al., 2017), timely disaster assessment (Gupta et al., 2019), and even strategic optimization of emergency responses (Zheng et al., 2021). In addition, such precise detection capabilities help track illegal construction activities (Suresh and Jain, 2013), assess the impact of policy changes, and support sustainable development goals. Consequently, leveraging VHR RS imagery for building change detection (BCD) represents a significant advancement towards intelligent and resilient urban management practices.

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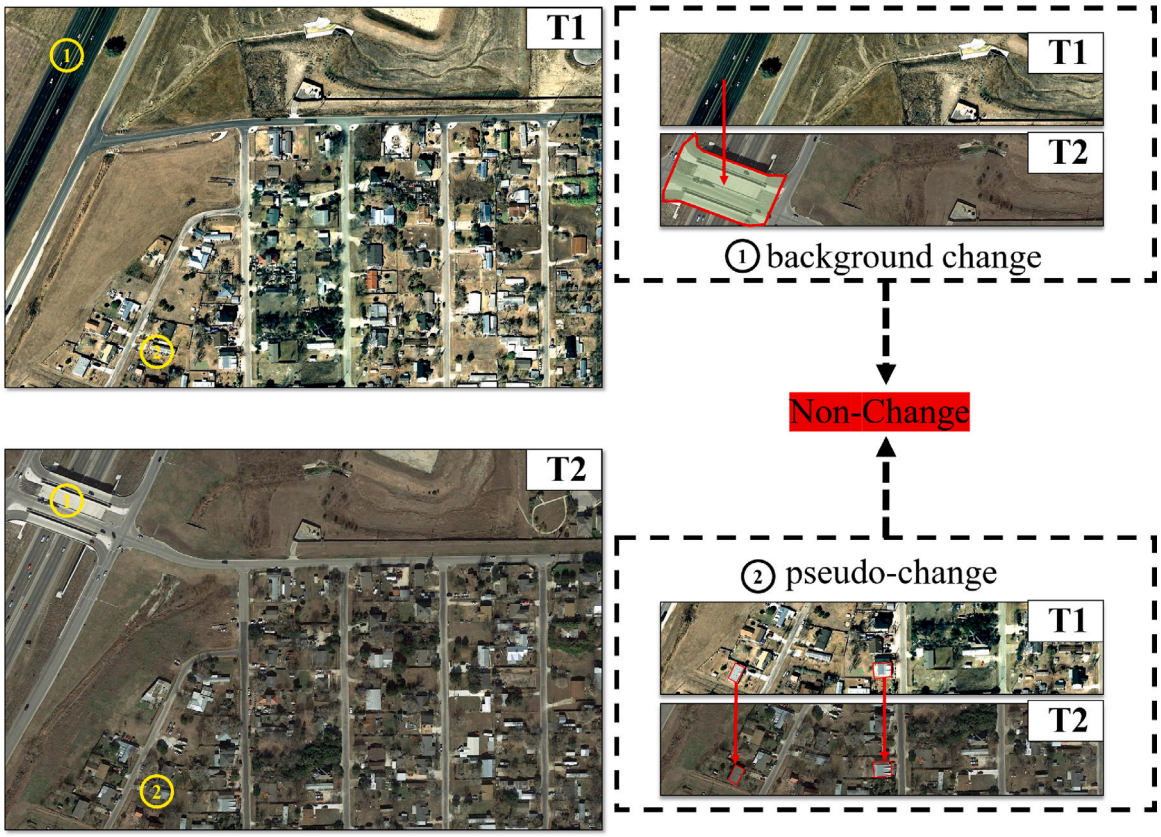


Fig. 1. Example of background change and pseudo-change in LEVIR-CD (Chen and Shi, 2020a).

In the past few decades, the research for change detection (CD) has witnessed significant advancements with the emergence of numerous approaches. Initially, various traditional techniques were proposed to tackle this complex task. Manual interpretation (Schwert et al., 2013), though reliable owing to its reliance on expert knowledge, proved to be both time consuming and labor intensive (Li et al., 2022b, 2024c). Subsequently, algebraic operations and transformations were introduced as alternative methods. Techniques such as image differencing (Quarmby and Cushnie, 1989), image regression (Ludeke et al., 1990), image rationing (Rignot and Van Zyl, 1993), and change vector analysis (Chen et al., 2003), were developed to identify changed regions effectively from bi-temporal data. Data reduction methods also played a crucial role in suppressing correlated information and emphasizing variance in input images. Methods such as principal component analysis (PCA) (Fung and LeDrew, 1987; Nielsen and Canty, 2008), tasseled cap transformation (KT) (Han et al., 2007), multivariate alteration detection (MAD) (Nielsen et al., 1998), Gram–Schmidt (GS) (Rosa et al., 2015), and chi-square (Vázquez-Jiménez et al., 2017) were instrumental in refining data for accurate CD. Furthermore, comparative classification (Hasanlou and Seydi, 2018), multiple classifications (Im and Jensen, 2005), and direct classification (Lu et al., 2004) techniques were employed, each offering a unique approach to enhance the detection process. However, these traditional methods often struggle to handle atmospheric conditions, seasonal variations, inconsistencies in satellite sensors, and variations in solar elevations (Han et al., 2023a). Consequently, there is a pressing need to develop more robust and adaptive techniques to overcome these limitations.

Owing to the remarkable advancements in computer vision technology, the research scope of CD has shifted towards embracing deep learning (DL) techniques. Convolutional neural networks (CNNs) (Krizhevsky et al., 2012) stand as prime examples of this trend, propelling CD to higher precision and efficiency. Subsequently, the emergence of the fully convolutional network (FCN Long et al. (2015)) marked a revolution in the field, introducing an innovative encoder–decoder architecture. Currently, FCN-based CD methods can be broadly categorized into two mainstreams: single-stream and double-stream. Single-stream CD methods adopt an early-fusion strategy, directly integrating bi-temporal VHR images and then processing through standard FCN networks to yield the final CD result. For example, Peng et al. (2019) directly concatenated images from two distinct periods and used a modified UNet++ (Zhou et al., 2018) to generate the final change map. However, this single-stream framework is susceptible to information-level confusion, which could lead to a notable loss of critical change information between the bi-temporal images (Zhang et al., 2020a).

In contrast with single-stream CD methods, double-stream strategies begin with the use of a Siamese encoder to extract features from bi-temporal images (Li et al., 2022a; Cheng et al., 2024). Subsequently, various decoding strategies are employed to effectively aggregate these encoded features from the two streams and generate a precise change map. Regarding Siamese encoders, some researchers directly utilize pre-trained backbone networks from the extensive ImageNet dataset (Deng et al., 2009), including prominent networks such as VGG (Simonyan and Zisserman, 2014), ResNet (He et al., 2016), ResNext (Xie et al., 2017), Res2Net (Gao et al., 2019), and transformer-based networks such as Swin Transformer (Liu et al., 2021) and pyramid vision transformer (PVT Wang et al. (2021a)). In addition, some researchers develop specialized encoders tailored to specific requirements. For instance, Chen et al. (2022b) introduced a fully convolutional Siamese network to extract multi-scale features, while Lin et al. (2022) approached CD as a video understanding problem, creating separate spatial and temporal encoders to capture spatial and temporal cues, respectively. In terms of decoding strategies, the focus is on extracting change-aware information from dual-stream encoders and producing a highly accurate change map. For example, Shu et al.

(2022) emphasized the integrity of change by considering the relationships and similarities between each pixel and its contextual pixels in difference-aware features. Liu et al. (2023) employed attention and transformer modules to model contextual information in bi-temporal images, introducing feature exchange mechanisms to bridge the domain gap between different temporal image domains. Ding et al. (2022) integrated temporally semantic features into the deep CD unit, enabling reasoning about bi-temporal semantic correlations and achieving precise CD results. In general, advances in decoding strategies underscore the importance of extracting and aggregating change-aware information for accurate CD. These developments highlight the ongoing progress towards robust and sophisticated CD techniques.

Recently, uncertainty theory has been extensively explored in computer vision, particularly in segmentation and detection tasks for natural and multi-modal images (Fang et al., 2023; Zhang et al., 2023; Yang et al., 2021; Baumgartner et al., 2019; Zhang et al., 2020b). Uncertainty is generally categorized into two types: aleatoric uncertainty, which arises from inherent noise and variability in data, and epistemic uncertainty, which stems from model's limitations or incomplete knowledge. These approaches often leverage Bayesian techniques and GANs to model data-level and model-level uncertainties. In RS, similar concepts, such as using Bayesian networks for hyperspectral image processing (Alcolea and Resano, 2022) and uncertainty-aware methods for building extraction (Li et al., 2024a) have been adopted. However, in the field of CD, existing work that addresses uncertainty has primarily focused on two key issues: (1) class imbalance between changed and unchanged regions and (2) difficulties in identifying object boundaries, that is edge-related uncertainties often degrade model performance. For example, Xu et al. (2024) introduced a dynamic adaptive focal loss to mitigate class imbalance, while Wang et al. (2021b) proposed an enhanced focal contrast loss to address similar issues. Additionally, Yin et al. (2023) developed an attention-guided attention network designed to improve boundary accuracy and reduce uncertainties near edges and small targets. In contrast, our work provides a more thorough exploration and categorization of uncertainty in CD.

First, current BCD datasets are primarily designed to detect changes in buildings, with changes in other land-cover classes typically classified as unchanged background. For example, datasets like LEVIR-CD (Chen and Shi, 2020a) may overlook transitions such as a road being converted into a bridge, treating these as background changes (Fig. 1). Because of the manual nature of labeling, background elements that resemble buildings, such as roads and bridges, are at significant risk of being misclassified as building changes. This mislabeling introduces aleatoric uncertainty at the data level, complicating both training and inference for models. Second, BCD tasks are complicated by pseudo-changes (Fig. 1), that is alterations in the appearance, texture, or style of buildings do not correspond to actual semantic changes. Combined with the intricate distribution of ground objects and severe class imbalance, these factors lead to high model-level epistemic uncertainty when identifying challenging samples.

Our approach systematically addresses the different types of uncertainty in BCD tasks, with a focus on the specific challenges posed by RS data. Unlike previous studies that focused mainly on general uncertainty modeling, we delve deeper into the unique uncertainties inherent in the CD process. Factors such as labeling errors, pseudo-changes, and the complex distribution of ground objects contribute to uncertainty in BCD, limiting the effectiveness of existing methods.

To address these challenges, we introduce the Uncertainty-Aware BCD Framework (UA-BCD). Our approach first employs a Siamese PVT (Wang et al., 2022) as an encoder to extract and aggregate multi-level features from bi-temporal images. A general decoding strategy is then used to produce an initial coarse CD map, along with its associated uncertainty. To specifically tackle aleatoric uncertainty, we present an aleatoric uncertainty estimation module (AUEM), which leverages conditional variational autoencoder (CVAE) technology to accurately estimate and constrain the consistency between the prior and posterior distributions of noise, effectively quantifying aleatoric uncertainty at the data level. To further enhance the aggregated multi-level features, we embed the estimated aleatoric uncertainty into the feature space and introduce a knowledge-guided feature enhancement module (KGFEM) to leverage the knowledge encoded in the coarse CD map, distinguishing between changed and unchanged regions. The refined features are subsequently decoded to produce a refined CD map. Lastly, we design an epistemic uncertainty estimator (EUE), which takes the concatenated bi-temporal images and the refined CD map as input and estimates the epistemic uncertainty. Supervised by the entropy calculated from the refined CD map, our UA-BCD can generate a CD map with lower epistemic uncertainty, indicating a higher degree of confidence.

The main contributions of this study are as follows.

1. We introduce the concept of uncertainty in BCD and propose the Uncertainty-Aware BCD (UA-BCD) Network to address these challenges.
2. We propose three innovative modules: the AUEM to model data-level uncertainty, the KGFEM to enhance feature representation using coarse change maps, and the EUE to manage model-level uncertainty.
3. We validate the effectiveness of the UA-BCD network through extensive experiments on five public BCD datasets. Furthermore, we assess its potential for large-scale applications by conducting a comprehensive study with a 5-year interval (2014–2024) on building changes in the Dongxihu District of Wuhan.

2. Methodology

2.1. Overview

To solve inherent uncertainty problems in BCD, we introduce an UA-BCD Framework. Within this framework, the training dataset comprises triplets (A, B, Y), where A represents the pre-change image, B signifies the post-change image, and Y denotes the corresponding change label.

As illustrated in Fig. 2, the process is divided into four stages. First, we adopt a general decoding strategy to generate an initial coarse CD map, albeit with relatively high uncertainty. Second, the AUEM is used to quantify aleatoric uncertainty at the data level. Third, we integrate the estimated aleatoric uncertainty into the feature space and propose a KGFEM to enhance the aggregated multi-level features and output a refined CD map. Finally, we introduce an independent EUE to produce an estimate of epistemic uncertainty, supervised by the entropy derived from the refined CD map. Using EUE, the UA-BCD framework can generate a change map with reduced epistemic uncertainty, reflecting a higher confidence level. The following sections will provide a detailed description of each of these four stages.

2.2. Coarse change map generation

Given the complexities and contextual nuances of RS imagery, transformer architectures, equipped with self-attention mechanisms, excel at capturing global dependencies, often outperforming CNNs in this domain. Therefore, we employ the PVT-2-b2 (Wang et al., 2022) as our feature extractor to independently encode multi-layer features from both pre-change and post-change images. This process is expressed as follows:

$$\begin{aligned} E_A^i &= Encoder(A), \quad i = 1, 2, \ldots, 4, \\ E_B^i &= Encoder(B), \quad i = 1, 2, \ldots, 4. \end{aligned} \tag{1}$$

Here, E_A^i and E_B^i represent the pre-change and post-change encoded features, respectively.

Next, we introduce an aggregation unit to combine the bi-temporal features at each level and use a feature pyramid network (FPN Lin et al.

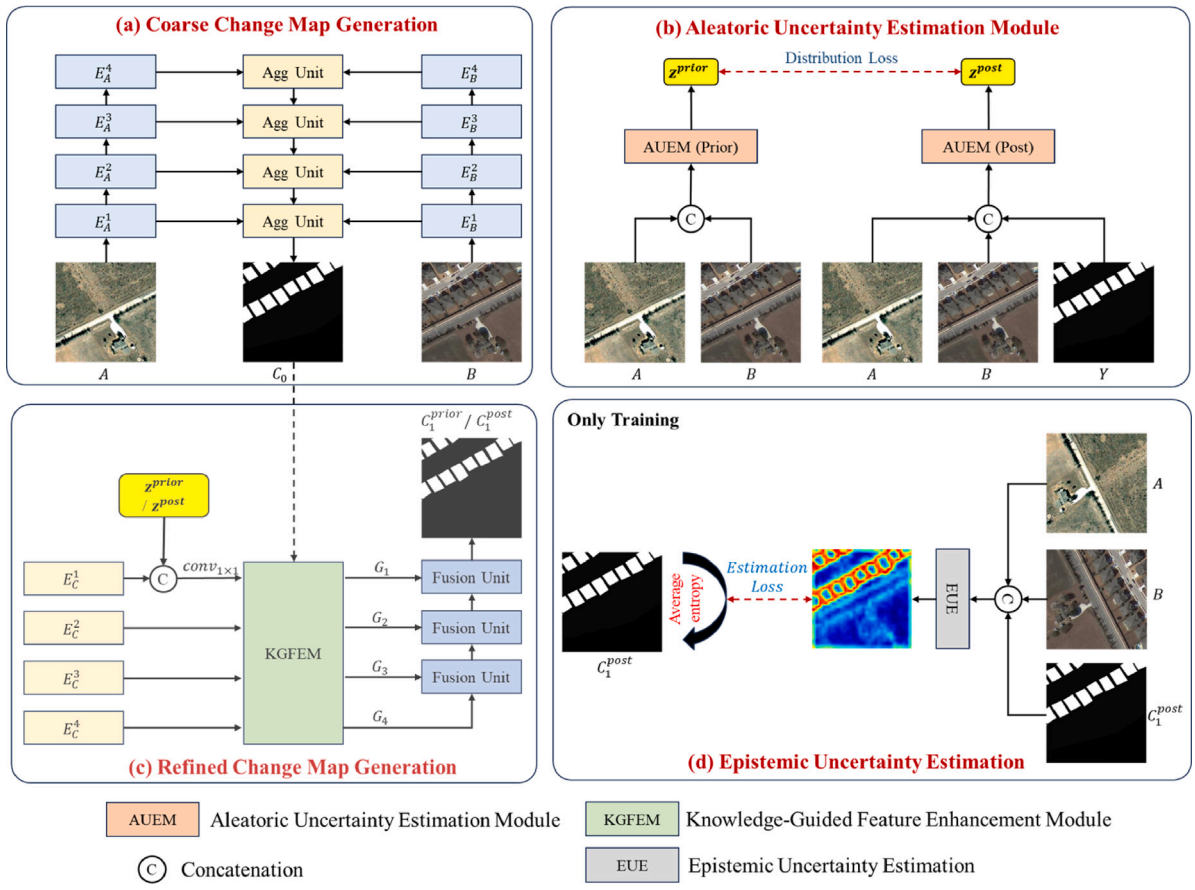


Fig. 2. Structure of the UA-BCD framework, where AUEM, KGFEM, and EUE are short for Aleatoric Uncertainty estimation module, Knowledge-guided feature enhancement module, and Epistemic uncertainty estimator, respectively.

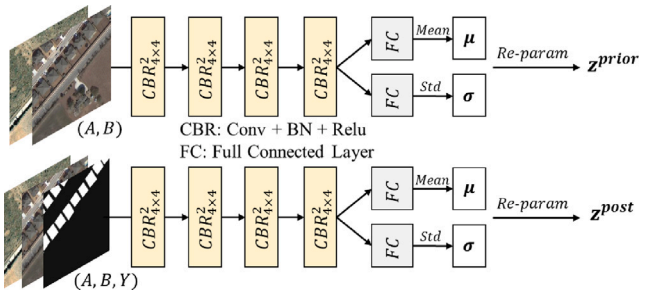


Fig. 3. Detailed structure of the Aleatoric uncertainty estimation module. In this diagram, $CBR^{m}_{k\times k}$ indicates a convolutional layer with a $k\times k$ kernel and a stride of m, and Re-param defines the Re-parameterization strategy.

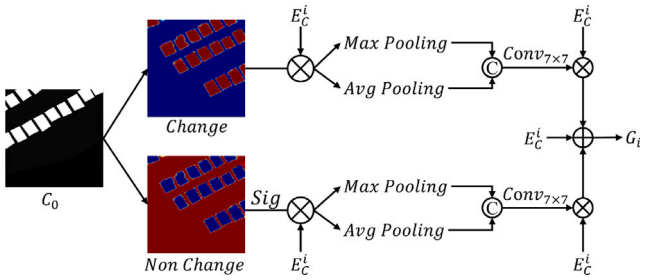


Fig. 4. Detailed structure of the Knowledge-guided feature enhancement module.

(2017)) to fuse them, generating an initial coarse CD map C_0 with high uncertainty. The entire process can be formulated as:

$$\begin{aligned} E_C^i &= Conv_{1\times1}(Concat(E_A^i, E_B^i)), \quad i = 1, 2, \ldots, 4, \\ T_4 &= Conv_{3\times3}(E_C^4), \\ T_i &= Conv_{3\times3}\left(U^2\left(Conv_{3\times3}(T_{i+1})\right) + E_C^i\right), \quad i = 1, 2, 3, \\ C_0 &= Conv_{1\times1}\left(Conv_{3\times3}(T_1)\right), \end{aligned} \tag{2}$$

where U^2 denotes an upsampling operation with a scaling factor of 2.

2.3. Aleatoric uncertainty estimation module

In the CD task, the primary objective is to minimize the overall loss function, which quantifies the model's performance. The loss function is formally expressed as follows:

$$Loss = \min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}\left(f(a_i, b_i; \theta), y_i\right), \tag{3}$$

where θ represents the model parameters of $f(\cdot)$, and (a_i, b_i, y_i) are individual samples drawn from the joint data distribution $p(A, B, Y)$. Here, A and B denote the input variables, typically images from different time periods, while Y represents the corresponding ground truth for CD. The function L measures the discrepancy between the model's predictions and the ground truth.

However, annotation errors in the training dataset inevitably introduce aleatoric uncertainty at the data level, complicating the CD task. To address this challenge, we propose the Aleatoric Uncertainty Estimation Module (AUEM), which leverages a conditional variational autoencoder (CVAE) to model the inherent data uncertainty. By using probabilistic inference, the CVAE allows us to estimate both the prior

and posterior distributions of latent variables, enabling the network to capture complex data variations and handle the uncertainty stemming from noisy or inconsistent annotations.

In the CVAE framework, the prior distribution of latent variables is conditioned on the input data and follows a Gaussian distribution. The conditional generative model in this study consists of three key components: conditional variables (A, B) (bi-temporal image pairs), latent variables z, and output variables Y. The latent variable is defined by the conditional distribution $P_\theta(z|A, B)$. Given the joint input conditional variable (A, B), the distribution of the output variable Y is given by $P_\omega(Y|A, B, z)$. Additionally, the posterior distribution of z is represented as $Q_\phi(z|A, B, Y)$. The CVAE loss function, designed to align the learned latent space with the true prior distribution, is given by:

$$L_{CVAE} = \mathbb{D}_{\mathrm{KL}}(Q_\phi(z|A, B, Y) \parallel P_\theta(z|A, B)), \tag{4}$$

where $\mathbb{D}_{\mathrm{KL}}(Q_\phi(z|A, B, Y) \parallel P_\theta(z|A, B))$ quantifies the divergence between the learned posterior distribution $Q_\phi(z|A, B, Y)$ and the prior distribution $P_\theta(z|A, B)$ using the Kullback–Leibler divergence. By minimizing this loss function, we aim to achieve a well-regularized latent space, enabling robust CD even in complex environments.

As illustrated in Fig. 3, AUEM consists of both prior and posterior networks designed to estimate aleatoric uncertainty. Initially, multiple inputs are concatenated and processed through four 4×4 convolutional blocks with a stride of 2 to extract hierarchical features while reducing spatial dimensions. These features are then flattened into a two-dimensional vector. This vector is mapped to a predefined dimensionality K (set to 8) using two parallel fully connected layers, producing the mean μ and standard deviation σ^2. The Gaussian distribution $N(\mu, \sigma^2)$ is then defined by these parameters, and latent variables are sampled as $z_k = \sigma_k \odot \epsilon + \mu_k$, where $\epsilon \in N(0, I)$. Here, z represents the aleatoric uncertainty, with the latent variables from the prior and posterior distributions denoted as z^{prior} and z^{post}, respectively.

The posterior is informed by the true change labels, guiding the model in learning aleatoric uncertainty at the data level. During the inference phase, we leverage the aleatoric uncertainty estimated by the prior component in conjunction with subsequent modules to achieve refined CD results, effectively mitigating data-level uncertainties without requiring the label map as input.

2.4. Refined change map generation

To integrate aleatoric uncertainty into the model learning process, we expand both z^{prior} and z^{post} into feature maps that match the spatial dimensions of the highest-level feature map E_C^4. Specifically, this aleatoric uncertainty, represented by z (where z can be either z^{prior} or z^{post}), is a latent variable generated by the hidden layer network. We concatenate this uncertainty with the highest-level aggregated feature E_C^4. To effectively embed the aleatoric uncertainty into the decoder, we then apply a 1×1 convolution to the concatenated feature map, restoring it to the original channel count of E_C^4.

Furthermore, based on the initial CD result C_0 from the previous stage, we apply a sigmoid function to obtain a probability map P, which indicates the probability of change occurrence for each pixel. To enhance multi-level features, we introduce a Knowledge-Guided Feature Enhancement Module (KGFEM) that utilizes the information embedded in C_0. The process is summarized as follows:

$$\begin{aligned} E_C^4 &= Conv_{1\times1}([E_C^4, z]), \\ G_i &= KGFEM(E_C^i, P), \end{aligned} \tag{5}$$

where $[\cdot]$ denotes the concatenation operation.

For KGFEM, as depicted in Fig. 4, we begin by downsampling C_0 to match the spatial size of E_C^4. We then apply a sigmoid function to derive two probability maps: P_c^4 and P_{nc}^4. In P_c^4, pixels with values close to 1 indicate a high probability of change occurrence, while in P_{nc}^4, pixels with values close to 1 indicate a high probability of non-change occurrence. By performing element-wise multiplication with P_c^4 and P_{nc}^4, we obtain change-aware features G_c^4 and non-change-aware features G_{nc}^4. This process can be expressed as:

$$\begin{aligned} &P_c^4 = Sig(D(C_0)), P_{nc}^4 = 1 - Sig(D(C_0)), \\ &G_c^4 = P_c^4 \cdot E_C^4, G_{nc}^4 = P_{nc}^4 \cdot E_C^4, \end{aligned} \tag{6}$$

where Sig is the sigmoid activation function, and D represents the downsampling operation.

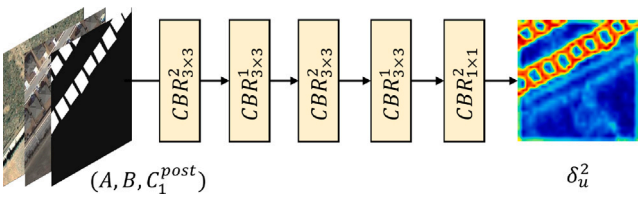


Fig. 5. Detailed structure of the epistemic uncertainty estimator.

Considering that the coarse CD result C_0 captures the correlation between each pixel's spatial position and its change/non-change status, we developed an implicit spatial attention mechanism to maximize the utilization of this information for feature enhancement. Our approach involves parallel processing, in which one branch identifies the maximum response value for each position across feature channels, capturing the most salient features, while the other branch averages the response values for each position across all channels, highlighting overall trends. These two spatially focused perception maps are then combined through a 7×7 convolutional layer, resulting in a unified spatial attention map that encapsulates both local and global spatial information. This attention map is activated through a sigmoid function, effectively modulating the input features to emphasize change/non-change information. Mathematically, the process is described as:

$$\begin{aligned} M_c^4 &= Sig(Conv_{7\times7}([Max(G_c^4), Avg(G_c^4)])), \\ M_{nc}^4 &= Sig(Conv_{7\times7}([Max(G_{nc}^4), Avg(G_{nc}^4)])), \\ G_4 &= M_c^4 \cdot E_C^4 + M_{nc}^4 \cdot E_C^4 + E_C^4, \end{aligned} \tag{7}$$

where Sig is the sigmoid activation function, $[\cdot]$ denotes the concatenation operation, and Max and Avg represent maximum and average pooling operations, respectively.

Utilizing this KGFEM, we enrich E_C^3 to produce G_3, similarly enhance E_C^2 to yield G_2, and refine E_C^1 to yield G_1. Finally, a general decoding strategy, specifically FPN (Lin et al., 2017), is employed to interpret and predict the final CD outcome C_1 with reduced uncertainty, which is formulated as:

$$\begin{aligned} N_4 &= Conv_{3\times3}(G_4), \\ N_i &= Conv_{3\times3}(U^2(Conv_{3\times3}(N_{i+1})) + N_i), \quad i = 1, 2, 3, \\ C_1 &= Conv_{1\times1}(Conv_{3\times3}(N_1)), \end{aligned} \tag{8}$$

where U^2 denotes an upsampling operation with a scaling factor of 2.

For both types of input uncertainty—prior uncertainty and posterior uncertainty—we carry out the previously mentioned decoding operations (refined CD generation), producing the corresponding outputs (C_1^{prior} and C_1^{post}). Importantly, during the inference phase, we concentrate exclusively on the learned prior aleatoric uncertainty to generate the final predictions, represented by C_1^{prior}.

2.5. Epistemic uncertainty estimation

By performing the aforementioned operations, we have made significant strides in estimating the distributions of both model parameters and prediction outcomes. This progress has laid a solid foundation for addressing epistemic uncertainty, which is a fundamental aspect of a model's inability to definitively predict changes.

Algorithm 1 Training Procedure of the UA-BCD Framework

Input: Training image pairs $\{a_i, b_i\}_n$ with corresponding ground truth labels $\{y_i\}_n$, maximum learning iterations M, batch size bs.

Output: Parameters θ for the generation of coarse change map, π for the generation of prior refined change map, ϕ for the generation of posterior refined change map, and ω for EUE.

Initialize θ, π, ϕ, and ω.
for $t = 1$ to M **do**
  Randomly sample a batch of image-label pairs $\{(a_i, b_i, y_i)\}_{bs}$ from the training set.
  for all (a_i, b_i, y_i) in the current batch **do**
    Obtain the coarse change map C_0 and the multi-level aggregated features E_C^i.
    Sample the prior $z_{prior}^i \sim p_\pi(z|(a_i, b_i))$ and posterior $z_{post}^i \sim p_\phi(z|(a_i, b_i), y_i)$.
    Generate the prior refined change map C_1^{prior} using the coarse change map C_0, the multi-level aggregated features E_C^i, and the estimated prior aleatoric uncertainty z^{prior}.
    Obtain the posterior refined change map C_1^{post} using the coarse change map C_0, the multi-level aggregated features E_C^i, and the estimated posterior aleatoric uncertainty z^{post}.
    Compute the posterior-based mean prediction μ_{post}^i by averaging the T samples.
    Calculate the epistemic uncertainty σ_p^2 on the basis of the posterior samples.
    Acquire the estimated epistemic uncertainty σ_u^2 using EUE.
  end for
  Jointly update the parameters θ, π, ϕ, and ω using the loss function defined in Eq. (11).
end for

This type of uncertainty primarily stems from variations in the sample distribution, reflecting the inherent complexities and nuances in the model's comprehension. To quantitatively capture this uncertainty, we define the mean outcome $\mu(Y|A, B)$ as the average prediction of our model, mathematically expressed as:

$$\mu(Y|A, B) = \frac{1}{T}\sum_{t=1}^{T} p(Y|A, B; \theta_t, \phi), \tag{9}$$

where T denotes the total number of samples considered for estimation, and $p(Y|A, B; \theta_t, \phi)$ represents the probability distribution of prediction Y given inputs A, B and model parameters θ_t and ϕ. Furthermore, the uncertainty associated with these predictions can be accurately gauged by calculating the entropy of this mean outcome. Accordingly, we define the variance of the mean model prediction as follows:

$$\sigma_p^2 = -\mu(Y|A, B)\log\mu(Y|A, B). \tag{10}$$

To effectively model and mitigate epistemic uncertainty, we introduce a specialized EUE. As depicted in Fig. 5, the architecture processes input by concatenating bi-temporal images A, B with the refined CD map C_1^{post}. It consists of four 3×3 convolutional blocks, each fortified with batch normalization and LeakyReLU activation, enhancing the model's ability to effectively represent the data. For improved computational efficiency, we apply a stride of 2 in the first and third convolutional layers, and in the final 1×1 convolutional layer during output generation. This strategy allows for the precise determination of the final uncertainty prediction result σ_u^2, accurately capturing the epistemic uncertainty inherent in our predictions.

To supervise and optimize the uncertainty estimation process, we use Eqs. (9) and (10) to calculate the average entropy across multiple prediction outcomes, with T set to 5. These entropy values are then sequentially normalized and binarized, serving as ground truth labels for the model. The learning of the EUE is guided by binary cross-entropy loss, which encourages continuous model improvement, reduces epistemic uncertainty, and ultimately improves the reliability and accuracy of our predictions.

2.6. Loss function

In stark contrast to previous CD networks, our innovative architecture incorporates four stages. This intricate setup necessitates the orchestration of multiple mutually independent optimization processes, as illustrated in Algorithm 1. Specifically, for the segmentation task, we rely on a straightforward binary cross-entropy loss function to ensure accurate pixel-level classification. To manage the distribution of prior and posterior aleatoric uncertainties generated by the AUEM, we employ the Kullback–Leibler divergence loss, effectively capturing the dissimilarity between the two distributions. Similarly, for epistemic uncertainty estimation, we harness binary cross-entropy loss to guide the model towards more reliable and precise uncertainty predictions.

Taken together, our comprehensive loss functions are formulated as follows:

$$\begin{aligned} L_{CD} =& BCE(U^4(C_0), Y) + BCE(BCE(U^4(C_1^{prior}), Y)) + \\ & BCE(U^4(C_1^{post}), Y) + \mathbb{D}_{\mathrm{KL}}(z^{post} \parallel z^{prior}), \\ L_{EUE} =& BCE(\sigma_u^2, \sigma_p^2). \end{aligned} \tag{11}$$

Here, U^4 indicates the upsampling operation with a scaling factor of 4, $BCE(C_0, Y)$ represents the binary cross-entropy loss between the coarse CD map C_0 and the ground truth Y. $BCE(C_1^{prior}, Y)$ and $BCE(C_1^{post}, Y)$ capture the loss associated with the prior and posterior outputs of the CVAE, respectively. The term $\mathbb{D}_{\mathrm{KL}}(z^{post} \parallel z^{prior})$ quantifies the divergence between the posterior and prior distributions in the latent space, ensuring a smooth and consistent representation of aleatoric uncertainty. $BCE(\sigma_u^2, \sigma_p^2)$ compares the estimated epistemic uncertainty σ_u^2 with the predicted variance σ_p^2, guiding the model towards more accurate uncertainty predictions.

Our optimization process consists of two parts. The first part optimizes Coarse Change Map Generation, AUEM, and Refined Change Map Generation based on L_{CD}. The second part optimizes the EUE on the basis of L_{EUE}. By combining these loss terms, we ensure that our model not only achieves superior segmentation performance but also provides reliable uncertainty estimates, thereby enhancing the overall reliability and interpretability of our change detection system.

3. Experiment

3.1. Dataset

To evaluate the performance of our proposed UA-BCD, we conducted extensive experiments on five large-scale BCD datasets.

1. LEVIR-CD Dataset (Chen and Shi, 2020a): This dataset consists of 637 pairs of RS images with a resolution of 0.5 m and a size of 1024×1024 pixels. It includes various styles of building change and is divided into training, validation, and test sets. For the experiments, we cropped the images into patches with a size of 256×256 pixels.
2. SYSU-CD Dataset (Shi et al., 2021): This dataset comprises 20,000 pairs of image tiles with a spatial size of 256×256 pixels, which defines various change categories such as buildings, roads, and vegetation. In this study, we concentrated on the change in buildings and used the pre-defined splits for training, validation, and testing.
3. Lebedev Dataset (Jiang et al., 2020): This dataset includes 16,000 pairs of RS images of 256×256 pixels, collected from Google Earth. It is divided into training, validation, and test sets.

4. WHU-CD Dataset (Ji et al., 2018): This dataset covers buildings in Christchurch, New Zealand, following an earthquake with a magnitude of 6.3 in February 2011. It includes large-scale images with a spatial size of $32,507 \times 15,354$ pixels and a resolution of 0.2 m/pixel. We partitioned the dataset into training, validation, and test sets with a cropping strategy.
5. Google-CD Dataset (Peng et al., 2020): This dataset contains 19 season-varying high-resolution image pairs of suburban areas in Guangzhou City, China, acquired between 2006 and 2019. The images have a spatial resolution of 0.55 m and vary in size from 1006 × 1168 pixels to 4936 × 5224 pixels, with annotations focused on buildings.

3.2. Implementation details

We performed non-overlapping cropping with a size of 256 × 256 pixels for all the datasets, except the LEVIR-CD dataset. For the LEVIR-CD dataset, we used overlapping cropping for the training set, with a crop size of 256 × 256 pixels and a stride of 128 pixels. For the validation and test sets of the LEVIR-CD dataset, we used non-overlapping cropping with a crop size of 256 × 256 pixels. Experiments on the SYSU-CD and Lebedev datasets followed the official train–test-validation splits. For the other two datasets, we adopted the same dataset partitioning method as Hierarchical Attention Network (HANet) (Han et al., 2023a). Our experiments were implemented with PyTorch 1.10 and Python 3.8 on an Nvidia GeForce RTX 4090 GPU with 24 GB of memory per card. During training, we used online data augmentation techniques such as random flipping, random rotation, and mosaic.

In the training process, optimization involved two parts. First, for the overall CD process, we used the Adam optimizer with an initial learning rate of 1×10^{-4}. Second, for the EUE, we also employed the Adam optimizer with the same initial learning rate of 1×10^{-4}. Both components were optimized using the same learning rate decay strategy, in which the learning rate was decayed by a factor of 10 every 50 epochs, and the training lasted for a total of 200 epochs.

3.3. Evaluation metrics

To conduct a comprehensive evaluation of our proposed model, we chose four metrics, *i.e.* , intersection over union (IoU), F1 score ($F1$), Precision (Pre), and Recall ($Recall$). First, we used TP, FP, and FN to represent the true positive, false positive, and false negative, respectively. Then, we defined the four evaluation metrics as follows:

$$IoU = \frac{TP}{TP + FP + FN}, \tag{12}$$

$$Precision = \frac{TP}{TP + FP}, \tag{13}$$

$$Recall = \frac{TP}{TP + FN}, \tag{14}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}. \tag{15}$$

3.4. Compared methods

To highlight the advantages of our proposed UA-BCD framework, we selected several state-of-the-art CD methods for qualitative and quantitative comparisons. Specifically, the comparison methods we selected include: Fully Convolutional Siamese-Concatenation (FC-Siam-conc) (Daudt et al., 2018), Fully Convolutional Siamese-Difference (FC-siam-diff) (Daudt et al., 2018), Spatial–Temporal Attention Neural Network (STANet) (Chen and Shi, 2020b), Siamese NestedUNet (SNUNet) (Chen et al., 2022a), Bitemporal Image Transformer (BIT) (Chen et al., 2021), ChangFormer (Bandara and Patel, 2022), Pair-to-Video Network (P2V-CD) (Lin et al., 2022), Hierarchical Attention Network (HANet) (Han et al., 2023a), Change Guiding Network (CGNet (Han et al., 2023b)), Multi-Scale Feature Fusion Change Detection Network (M-Swin) (Pan et al., 2024), and Bi-Temporal Adapter Network (BAN) (Li et al., 2024b).

Table 1
Performance comparison with baseline models on the LEVIR-CD dataset. ↑ indicates the higher score the better and vice versa. The best score for each metric is marked in bold. The second score for each metric is underlined.

Baseline	LEVIR-CD (%)			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
FC-Siam-conc	82.24	90.25	89.67	90.84
FC-Siam-diff	82.24	90.26	90.40	90.11
SNUNet	82.65	90.50	90.46	90.54
BIT	82.76	90.56	90.87	90.26
ChangeFormer	81.25	89.65	89.70	89.61
P2V-CD	83.67	91.11	91.01	<u>91.21</u>
HANet	82.27	90.28	91.21	89.36
CGNet	<u>85.21</u>	<u>92.01</u>	<u>93.15</u>	90.90
M-Swin	83.58	91.05	92.08	90.05
BAN	84.93	91.85	92.93	90.89
UA-BCD	**85.99**	**92.47**	**93.38**	**91.57**

3.5. Evaluation on the LEVIR-CD dataset

3.5.1. Quantitative comparison

As shown in Table 1, the comparison of UA-BCD with baseline models on the LEVIR-CD dataset highlights several key aspects of its performance. Specifically, UA-BCD achieves the highest IoU of 85.99%, a notable improvement over the nearest competitor, CGNet, which scores 85.21%. The F1 score of UA-BCD is 92.47%, again surpassing all other methods, with CGNet being the closest at 92.01%. This F1 score, which balances precision and recall, indicates that UA-BCD is more effective at accurately detecting changes while minimizing both missed changes and false detections. Moreover, UA-BCD records not only the highest precision (93.38%) but also the highest recall (91.57%). This balance is crucial, given that high precision ensures the correctness of detected changes, while high recall ensures that most actual changes are captured. In contrast, other methods often exhibit a slight drop in recall to maintain higher precision, reflecting a trade-off. However, UA-BCD excels in both areas, demonstrating its robustness across different scenarios. The enhancements in both IoU and F1 score, along with a balanced precision and recall, underscore the model's superior capability to address various challenges in RS image analysis, outperforming the current SOTA methods.

3.5.2. Visual comparison

To further demonstrate the superiority of our method, we conducted an extensive visual evaluation. As illustrated in Fig. 6, we selected two representative examples. In the first example, it is clear that SNUNet, BIT, and ChangeFormer struggle to detect any changed areas, with P2V also performing poorly. While CGNet and BAN manage to identify some of the changed buildings, our UA-BCD method delivers the best detection performance, closely matching the ground truth. In the second example, where a new building appears in the lower right corner between two time phases, none of the compared methods successfully detect the change. On the contrary, our proposed UA-BCD method perfectly identifies it. These examples vividly demonstrate the superiority of our method over existing CD techniques, underscoring the critical importance of addressing uncertainty to achieve precise and reliable results.

3.6. Evaluation on the SYSU-CD dataset

3.6.1. Quantitative comparison

As shown in Table 2, the UA-BCD method exhibits several notable advantages over baseline models.UA-BCD achieves the highest IoU score of 71.38%, surpassing the next best method, BAN, which scores 67.35%. This 4.03% improvement underscores UA-BCD's superior ability to accurately detect and segment changes in the SYSU-CD dataset. Additionally, UA-BCD excels with an F1 score of 83.30%, outperforming BAN, which scores 80.49%. This 2.81% enhancement highlights

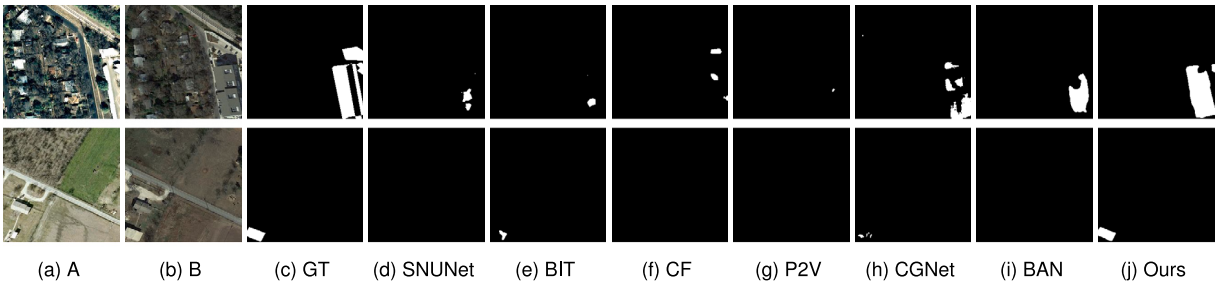


Fig. 6. Visual comparison on the LEVIR-CD dataset. CF is an abbreviation for ChangeFormer.

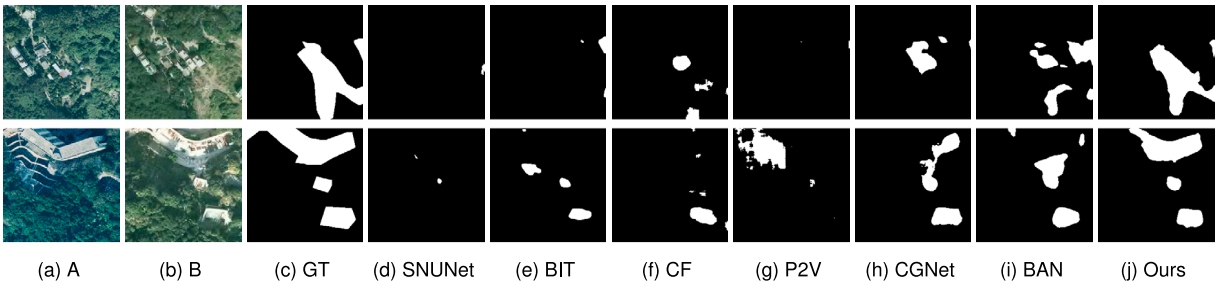


Fig. 7. Visual comparison on the SYSU-CD dataset. CF is an abbreviation for ChangeFormer.

Table 2
Performance comparison with baseline models on the SYSU-CD dataset.

Baseline	SYSU-CD (%)			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
FC-Siam-conc	66.16	79.64	83.27	76.31
FC-Siam-diff	66.93	80.19	79.96	**80.42**
SNUNet	64.79	78.63	80.88	76.50
BIT	65.33	79.03	78.23	79.84
ChangeFormer	65.24	78.97	81.21	76.84
P2V-CD	64.35	78.31	81.93	74.99
HANet	63.14	77.41	78.71	76.14
CGNet	66.55	79.92	86.37	74.37
M-Swin	66.44	79.90	87.46	73.55
BAN	67.35	80.49	**88.26**	73.97
UA-BCD	**71.38**	**83.30**	87.28	79.66

Table 3
Performance comparison with baseline models on the Lebedev dataset.

Baseline	Lebedev (%)			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
FC-Siam-conc	84.28	91.47	88.72	94.40
FC-Siam-diff	85.94	92.44	89.99	95.02
SNUNet	90.65	95.10	91.55	98.93
BIT	89.19	94.29	91.14	97.66
ChangeFormer	89.58	94.50	90.93	98.37
P2V-CD	91.14	95.36	91.51	**99.56**
HANet	77.51	87.33	92.33	82.85
CGNet	91.41	95.51	96.16	94.48
M-Swin	91.84	95.75	96.12	95.38
BAN	88.47	93.88	94.54	93.23
UA-BCD	**93.49**	**96.64**	**96.38**	96.90

UA-BCD's better balance between precision and recall, reflecting its effectiveness in accurately identifying changes while reducing both false positives and false negatives. Although BAN achieves the highest precision at 88.26%, UA-BCD's precision of 87.28% remains highly competitive. That is, while BAN slightly outperforms in precision, UA-BCD consistently delivers strong performance in detecting true changes. UA-BCD also leads in recall with a score of 79.66%, significantly higher than BAN's recall of 73.97%, marking a 5.69% improvement. This result demonstrates UA-BCD's superior capability in identifying all relevant changes in the dataset, ensuring that fewer actual changes are overlooked. Overall, UA-BCD demonstrates superior performance across all evaluated metrics compared with existing methods, particularly in IoU and F1 score. This finding highlights its effectiveness and robustness in CD tasks on the SYSU-CD dataset.

3.6.2. Visual comparison

Fig. 7 presents a visual comparative analysis using two representative examples from the SYSU-CD test set. In the first example, methods such as SNUNet, BIT, and P2V struggle to identify any changed areas, while ChangeFormer, CGNet, and BAN manage to detect only a small portion of the changes. By contrast, our proposed UA-BCD achieves segmentation results that are closest to the ground truth, effectively identifying the changes. In the second example, all compared methods fail to detect the largest changed area, with SNUNet missing all detected changes. Our UA-BCD, on the contrary, accurately identifies all three significant changed areas. These examples clearly demonstrate the superiority of our method and emphasize the importance of addressing uncertainty to enhance CD performance.

3.7. Evaluation on the Lebedev dataset

3.7.1. Quantitative comparison

Table 3 shows the performance comparison of various methods on the Lebedev dataset. Several key observations highlight the advantages of UA-BCD. Our UA-BCD method achieves the highest IoU score of 93.49%, surpassing the closest competitor, M-Swin, which scores 91.84%. This 1.65% improvement indicates that UA-BCD significantly enhances the accuracy of change detection and segmentation on the Lebedev dataset. In terms of the F1 score, UA-BCD leads with a value of 96.64%, well above the second-best method, M-Swin, which achieves 95.75%. This improvement of 0.89% underscores UA-BCD's superior balance between precision and recall, reflecting its enhanced capability in accurately detecting changes while minimizing both false positives and false negatives. Although P2V-CD has the highest recall at 99.56%, UA-BCD maintains a competitive recall score of 96.90%, which is notably higher than most other methods. This high recall score indicates that UA-BCD is effective at identifying all relevant changes, ensuring that fewer actual changes are missed. Regarding

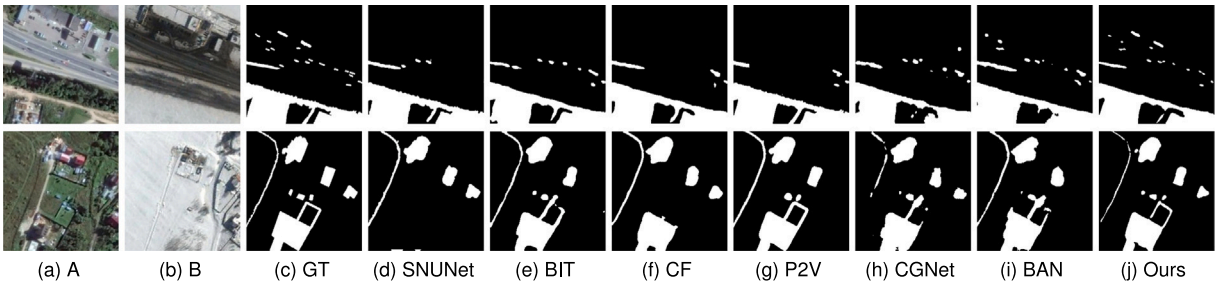


Fig. 8. Visual comparison on the Lebedev dataset. CF is an abbreviation for ChangeFormer.

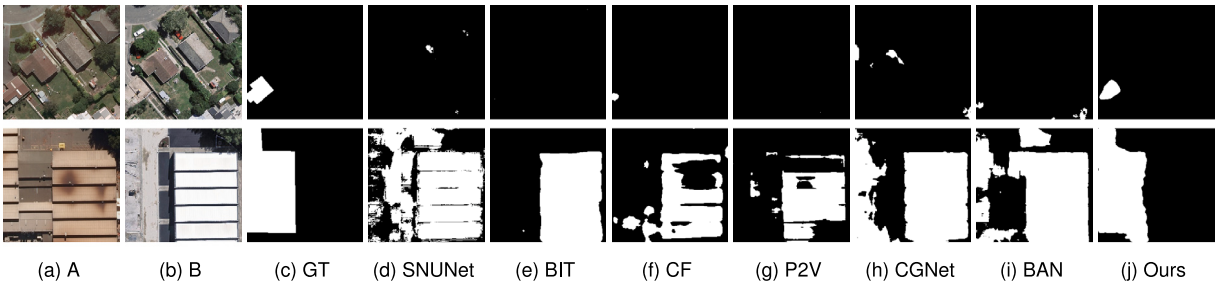


Fig. 9. Visual comparison on the WHU-CD dataset. CF is an abbreviation for ChangeFormer.

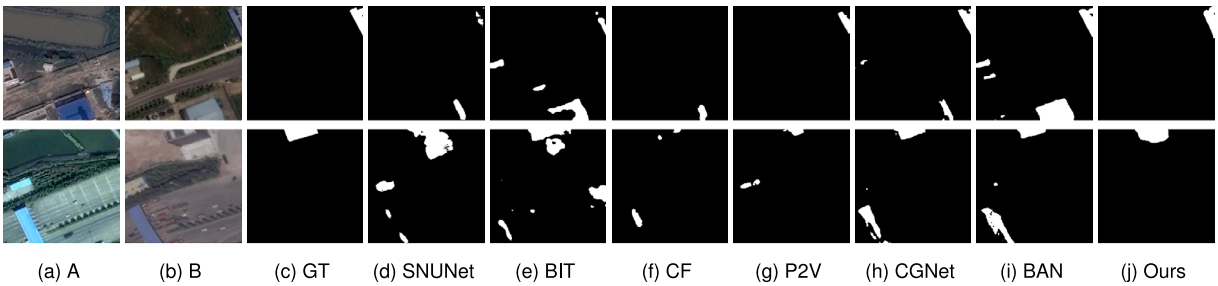


Fig. 10. Visual comparison on the Google dataset. CF is an abbreviation for ChangeFormer.

precision, UA-BCD achieves an impressive score of 96.38%, surpassing both CGNet (96.16%) and M-Swin (96.12%). This finding demonstrates that UA-BCD excels in detecting true changes while maintaining a high level of accuracy. Overall, UA-BCD demonstrates superior performance across all metrics compared with existing methods on the Lebedev dataset, particularly in IoU and F1 score. These results highlight the robustness and effectiveness of UA-BCD in handling CD tasks.

3.7.2. Visual comparison

For visual comparison, we selected two scenes from the Lebedev dataset test set, both showcasing various building changes, as illustrated in Fig. 8. In the first scene, the primary challenge is the small building changes in the upper area, which most of the compared methods can only partially detect. By contrast, our proposed UA-BCD successfully identifies nearly all of these changes. In the second scene, seasonal variations create significant difficulties in detecting all altered buildings. Among the compared methods, P2V-CD performs the best but still misses some changed areas. On the contrary, our UA-BCD achieves the highest overall performance. This underscores the effectiveness of our approach in identifying even the smallest alterations, highlighting the strength of our uncertainty-handling strategy.

3.8. Evaluation on the WHU-CD dataset

3.8.1. Quantitative comparison

As detailed in Table 4, UA-BCD significantly outperforms other methods on the WHU-CD dataset. It achieves the highest scores in all metrics, namely IoU, F1 score, precision, and recall. Specifically, UA-BCD attains an IoU of 86.87%, surpassing CGNet, the second best, by 0.66%. It also leads with an F1 score of 92.98%, which is 0.41% higher than BAN's score. Although CGNet achieves the highest precision at 94.47%, UA-BCD's precision of 94.71% is still highly competitive. Furthermore, UA-BCD excels in recall with a score of 91.30%, outperforming BAN by 0.15%. These results highlight UA-BCD's overall superior performance and robustness in effectively detecting and segmenting changes in the WHU-CD dataset.

Table 4
Performance comparison with baseline models on the WHU-CD dataset.

Baseline	WHU-CD (%)			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
FC-Siam-conc	40.89	58.05	76.49	46.77
FC-Siam-diff	47.05	63.99	72.06	57.55
SNUNet	78.19	87.76	87.84	87.68
BIT	68.02	80.97	74.01	89.37
ChangeFormer	77.27	87.16	92.70	82.28
P2V-CD	78.03	87.66	86.99	88.33
HANet	78.82	88.16	88.30	88.01
CGNet	86.21	92.59	94.47	90.79
M-Swin	80.85	89.41	90.39	88.45
BAN	86.05	92.57	94.03	91.15
UA-BCD	**86.87**	**92.98**	**94.71**	**91.30**

Table 5
Performance comparison with baseline models on the Google-CD dataset.

Baseline	Google-CD (%)			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
FC-Siam-conc	71.12	83.13	79.86	86.67
FC-Siam-diff	71.52	83.40	82.72	84.08
SNUNet	72.97	84.37	83.65	85.11
BIT	75.73	86.19	86.57	85.81
ChangeFormer	71.90	83.65	85.18	82.18
P2V-CD	74.04	85.08	87.12	83.14
HANet	73.14	84.49	86.03	83.00
CGNet	75.27	85.89	88.07	83.82
M-Swin	76.12	86.22	88.17	84.35
BAN	78.70	88.11	89.22	87.02
UA-BCD	**79.86**	**88.81**	**89.66**	**87.97**

3.8.2. Visual comparison

Fig. 9 showcases two representative visual examples that highlight the effectiveness of our method. In the first scenario, all comparison methods struggle to accurately identify the changed areas, with SNUNet, CGNet, and BAN even misclassifying some unchanged regions as changed. In contrast, our method successfully segments the changed areas with high precision. In the second scenario, while all methods incorrectly classify the building on the right side of the image as changed, only CGNet and BAN partially identify the changed areas, but their results lack structural consistency. Our method, on the contrary, accurately segments the building with changes, offering a more precise and complete detection. These examples clearly illustrate our method's superiority in reducing both misclassification and missed detection.

3.9. Evaluation on the google-CD dataset

3.9.1. Quantitative comparison

Table 5 presents the performance comparison of UA-BCD against baseline models on the Google-CD dataset. Notably, UA-BCD outperforms all other methods across all metrics. It achieves the highest IoU of 79.86%, surpassing the closest competitor, BAN, which scores 78.70%. This 1.16% improvement highlights UA-BCD's superior capability in accurately detecting and segmenting changes. In terms of the F1 score, UA-BCD leads with a value of 88.81%, which is higher than BAN's 88.11% by 0.70%. This indicates that UA-BCD provides a better balance between precision and recall, effectively minimizing false positives and false negatives. While BAN achieves a precision of 89.22%, UA-BCD's precision of 89.66% is slightly better, reflecting its effectiveness in accurately detecting changes while avoiding false positives. Additionally, UA-BCD's recall of 87.97% surpasses BAN's recall of 87.02% by 0.95%, showcasing its ability to capture more true changes without missing significant alterations. Overall, UA-BCD demonstrates superior performance across all evaluated metrics, confirming its robustness and effectiveness in CD on the Google-CD dataset compared with existing methods.

3.9.2. Visual comparison

Fig. 10 presents two visual examples highlighting the qualitative performance of our method. In the first scenario, SNUNet and ChangeFormer struggle to accurately segment the changed areas. While BIT, CGNet, and BAN are able to identify some changes, they suffer from substantial misclassification issues. P2V-CD, although avoiding misclassification, only captures a limited portion of the changed areas. By contrast, our method achieves the most precise and comprehensive segmentation. In the second scenario, all other methods display significant misclassification errors, whereas our method provides segmentation results that closely align with the ground truth. These examples underscore the effectiveness of our approach in reducing uncertainty and enhancing the accuracy of CD.

4. Ablation study

In this section, we provide a comprehensive and in-depth analysis of each module within UA-BCD. All experiments were conducted on the LEVIR-CD, SYSU-CD, and Google-CD datasets under consistent experimental conditions to ensure fairness.

4.1. Performance gains of AUEM

Incorporating AUEM substantially enhances model performance across all datasets, as detailed in Table 6. For LEVIR-CD, AUEM improves the Intersection over Union (IoU) by 1.69%, the F1 score by 1.00%, Precision by 0.32%, and Recall by 1.63%. In SYSU-CD, the IoU increases by 2.84%, the F1 score by 2.09%, Precision by 4.32%, and Recall by 1.12%. For Google-CD, AUEM leads to a rise in IoU by 1.70%, F1 score by 1.09%, and Recall by 2.94%. These improvements highlight AUEM's effectiveness in utilizing a CVAE to estimate uncertainty at the data level. By leveraging bi-temporal images to estimate prior uncertainty and integrating it with posterior uncertainty conditioned on labels, AUEM applies a distribution loss constraint that helps the model focus more effectively on significant changes while mitigating the impact of irrelevant background variations. This modification results in improved detection accuracy, allowing the model to better identify and prioritize substantial changes and thus enhancing overall feature extraction and detection precision.

4.2. Quantifying the benefits of KGFEM

To assess the impact of incorporating both changed and unchanged regions from the coarse CD map C_0, we conducted experiments comparing feature enhancement with and without this information. As shown in Table 6, integrating both changed and unchanged region information improves model performance across datasets: for LEVIR-CD, IoU increases by 1.87%, F1 score by 1.05%, Precision by 0.35%, and Recall by 1.71%; for SYSU-CD, IoU rises by 4.20%, F1 score by 3.64%, Precision by 4.99% and Recall by 2.57%; and for Google-CD, IoU improves by 2.39%, F1 score by 1.47%, and Recall by 3.37%. Additionally, the combined use of AUEM and KGFEM (A+ B+ C) results in even greater benefits, clearly demonstrating KGFEM's effectiveness.

In the KGFEM, we utilized the change areas reflected by C_0 along with the non-changing background information to guide subsequent feature enhancement. To explore the benefits of this combined guidance, we conducted comparative experiments. As illustrated in Table 7, using both change area and non-changing background information for feature enhancement consistently outperforms using only change area information across the three datasets. This underscores the importance of considering background context to enhance the model's understanding and accuracy in identifying changes.

Fig. 11 illustrates that features before KGFEM processing do not clearly capture change information. With guidance from C_0, there is a significant enhancement in features across all layers. This result demonstrates KGFEM's effectiveness, given that it balances local high-response region information with global context through max pooling and average pooling operations. Additionally, KGFEM integrates knowledge about changed and unchanged regions from C_0 into features at all levels using a non-explicit attention mechanism, which greatly refines feature extraction.

4.3. Impact study of EUE implementation

Table 6 demonstrates the substantial performance improvements resulting from the implementation of EUE across all datasets. On the LEVIR-CD dataset, EUE enhances IoU by 0.54%, the F1 score by 0.50%, Precision by 0.05%, and Recall by 0.93%. For the SYSU-CD dataset, IoU increases by 0.73%, F1 score by 0.87%, Precision by 0.79%, and Recall by 0.93%. In the Google-CD dataset, IoU rises by 0.10%, F1 score

Table 6
Ablation experiments on the test datasets. A represents the coarse change map generation. B, C, D represent AEUM, KGFEM, and EUE, respectively.

Setting				LEVIR-CD				SYSU-CD				Google-CD			
A	B	C	D	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
✓				83.59	91.06	92.25	89.91	63.73	77.84	82.54	73.65	76.23	86.51	88.64	84.48
✓	✓			85.28	92.06	92.57	91.54	66.57	79.93	86.86	74.77	77.93	87.60	87.77	87.42
✓		✓		85.46	92.11	92.60	91.62	67.93	81.48	87.53	76.22	78.62	87.98	88.11	87.85
✓			✓	84.13	91.56	92.30	90.84	64.46	78.71	83.33	74.58	76.33	86.67	88.70	84.74
✓	✓	✓		85.50	92.18	93.31	91.08	69.42	81.95	86.24	78.07	78.95	88.24	87.97	**88.50**
✓	✓	✓	✓	**85.99**	**92.47**	**93.38**	**91.57**	**71.38**	**83.30**	**87.28**	**79.66**	**79.86**	**88.81**	**89.66**	87.97

Table 7
Ablation experiments on the test datasets. C and NC are short for change and non-change, respectively.

Combination		LEVIR-CD				SYSU-CD				Google-CD			
C	NC	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑	*IoU* ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
✓		85.08	91.94	92.79	91.10	70.05	82.48	86.32	78.97	78.59	88.01	88.72	87.32
✓	✓	**85.99**	**92.47**	**93.38**	**91.57**	**71.38**	**83.30**	**87.28**	**79.66**	**79.86**	**88.81**	**89.66**	**87.97**

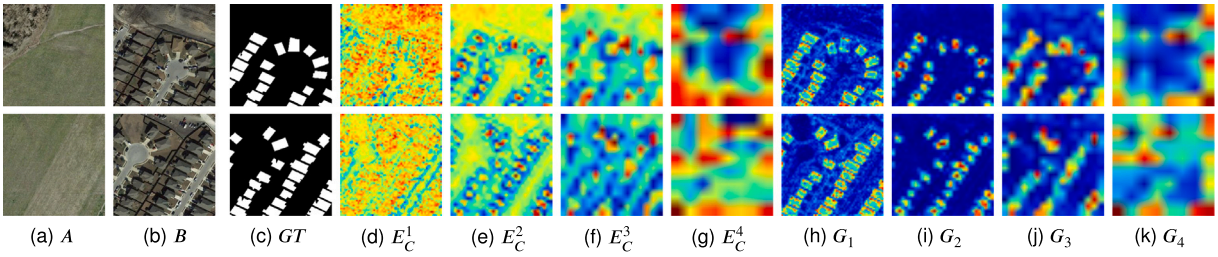


Fig. 11. Feature visual examples before/after KGFEM.

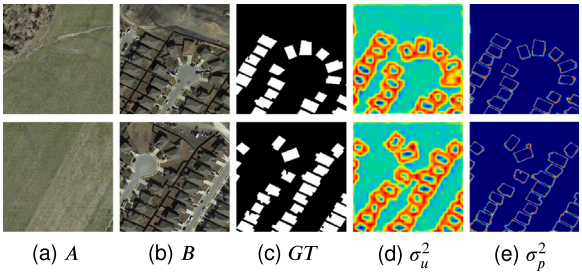


Fig. 12. Visual example of σ_u^2 and σ_p^2.

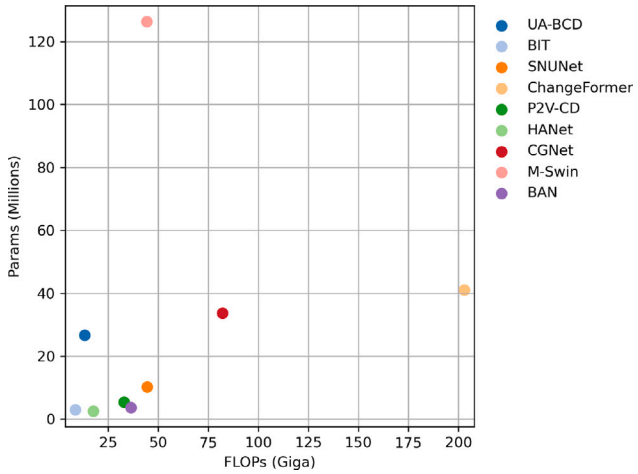


Fig. 13. Comparison of the complexity of our method with the SOTA methods for BCD.

by 0.16%, Precision by 0.06%, and Recall by 0.26%. Furthermore, the introduction of EUE, in conjunction with AUEM and KGFEM (A + B + C + D), enables the model to achieve optimal performance, effectively demonstrating the advantages of addressing epistemic uncertainty.

Fig. 12 visualizes the entropy values of segmentation results alongside the epistemic uncertainty estimated by EUE. The high-entropy areas, indicating regions of greater uncertainty, are minimal and predominantly located around the edges of changed regions. This finding highlights EUE's effectiveness in managing epistemic uncertainty and guiding the model to focus on regions with high uncertainty. While AUEM addresses aleatoric uncertainty by incorporating data-level uncertainty into feature decoding, EUE models cognitive uncertainty at the model level by estimating it from bi-temporal images and posterior predictions.

By supervising the entropy of posterior predictions, EUE encourages the decoding module to focus on high-uncertainty pixels, thereby enhancing cognitive performance. AUEM and EUE were co-trained, optimizing independently yet benefiting mutually from their complementary contributions. Importantly, EUE was not used during inference, which significantly reduces computational load.

4.4. Visual analysis

As illustrated in Fig. 14, we visually compared the ablation effects of each module across four challenging scenes. It is evident that the model's capacity to handle complex challenges significantly improves with the sequential integration of the AUEM and KGFEM, compared with the initial coarse decoding results. Despite this enhancement, the model still struggles with issues such as shadows and changes in building styles that introduce pseudo-changes. On the contrary, the introduction of the EUE module markedly enhances the final UA-BCD's ability to address these uncertainty challenges, resulting in CD outcomes that closely approximate the ground truth labels.

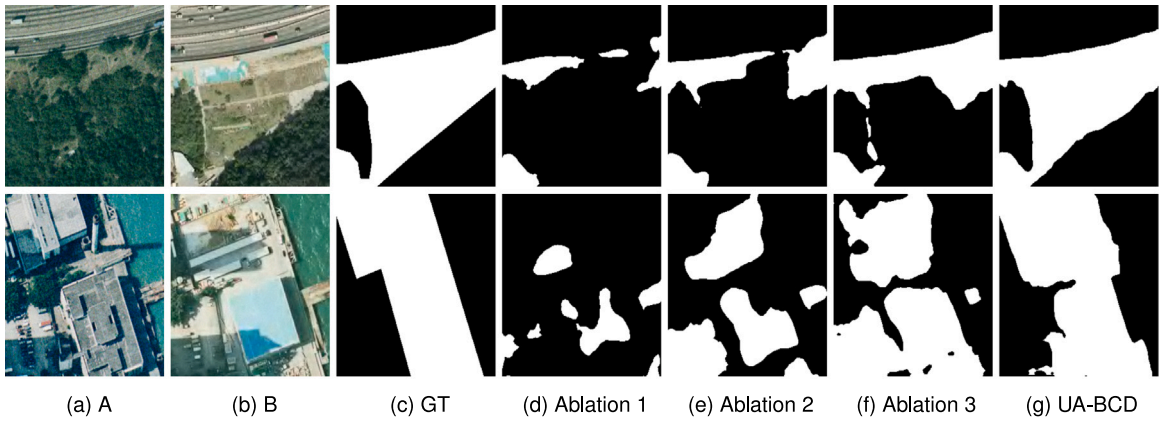


Fig. 14. Visual comparison of the ablation study results on the test datasets. Ablation 1, Ablation 2, and Ablation 3 refer to the Ablation settings A, A+ B, and A+ B+ C, respectively.

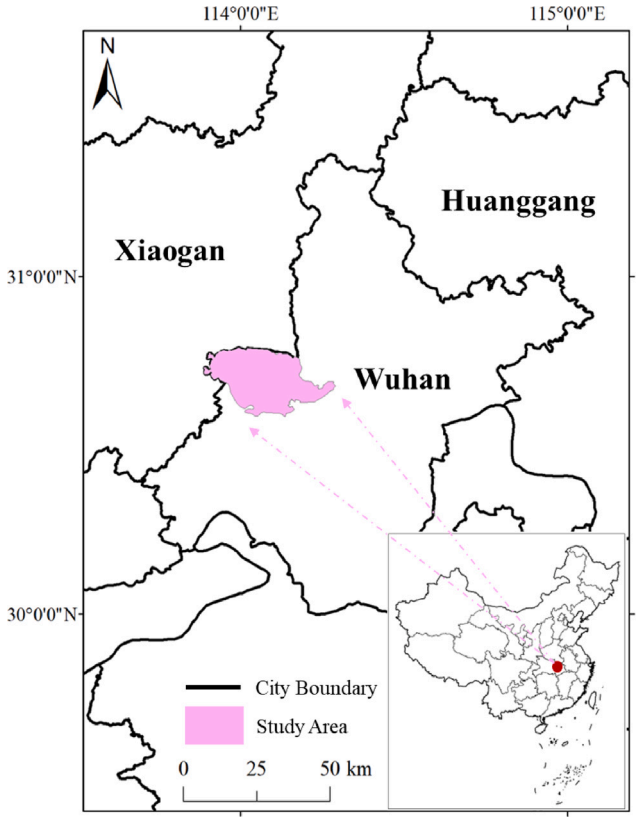


Fig. 15. Location of the study area, with the selected Dongxihu district, Wuhan city, highlighted in pink.

4.5. Complexity analysis

As shown in Fig. 13, we compared our UA-BCD method with other SOTA methods in terms of parameter count and FLOPs. Despite incorporating uncertainty estimation, our method maintains relatively low computational complexity, with a total parameter count of only 26.637 MB—much lighter than models like M-Swin and CGNet. This balance between accuracy and efficiency highlights the effectiveness of our approach in addressing uncertainty while keeping the model lightweight.

Table 8 further demonstrates that models utilizing explicit attention mechanisms, such as ChangeFormer, BAN, M-Swin, and UA-BAD, achieve notable accuracy gains by leveraging matrix operations to effectively model correlations. However, this enhancement results in higher computational costs, leading to slower inference times. Specifically, our UA-BCD model, which uses PVT-v2-b2 with shared weights for bi-temporal feature extraction, experiences slower inference primarily due to the self-attention mechanisms of the transformer architecture. Additionally, AUEM contributes to this slower speed, because modeling data-level uncertainty introduces extra computation. Nevertheless, the added complexity is compensated by the substantial improvements in accuracy and robustness, as the explicit attention mechanisms and uncertainty modeling allow for more precise handling of complex data. Although this trade-off results in longer inference times, the overall performance gains justify the additional computational cost.

Table 8
FPS comparison with baseline models.

Baseline	FPS	Inference time (s)
FC-Siam-conc	162.13	0.006168
FC-Siam-diff	**165.88**	**0.006028**
SNUNet	68.49	0.014600
BIT	131.17	0.007624
ChangeFormer	36.36	0.027506
P2V-CD	121.53	0.008229
HANet	96.20	0.010395
CGNet	98.09	0.010195
M-Swin	17.98	0.055614
BAN	27.22	0.036739
UA-BCD	29.34	0.034080

5. UA-BCD for large-scale urban building detection: A detailed case study

To explore the potential of our proposed method for BCD in large urban areas under real-world scenarios, we conducted a detailed application validation in this section.

5.1. Study area

Wuhan, the core city in central China, has experienced rapid urban development over the past decade. Among its districts, Dongxihu District is one of the areas that have undergone the most significant changes. Over the last 10 years, numerous technology companies have established themselves in this area, leading to the construction of many high-rise buildings. Therefore, in this study, we selected Dongxihu District as our research area (as shown in Fig. 15), focusing on building changes from 2014 to 2024, with a five-year interval for analysis.

We obtained RGB optical images of Wuhan's Dongxihu District for the years 2014, 2019, and 2024 from Google Earth, with a resolution

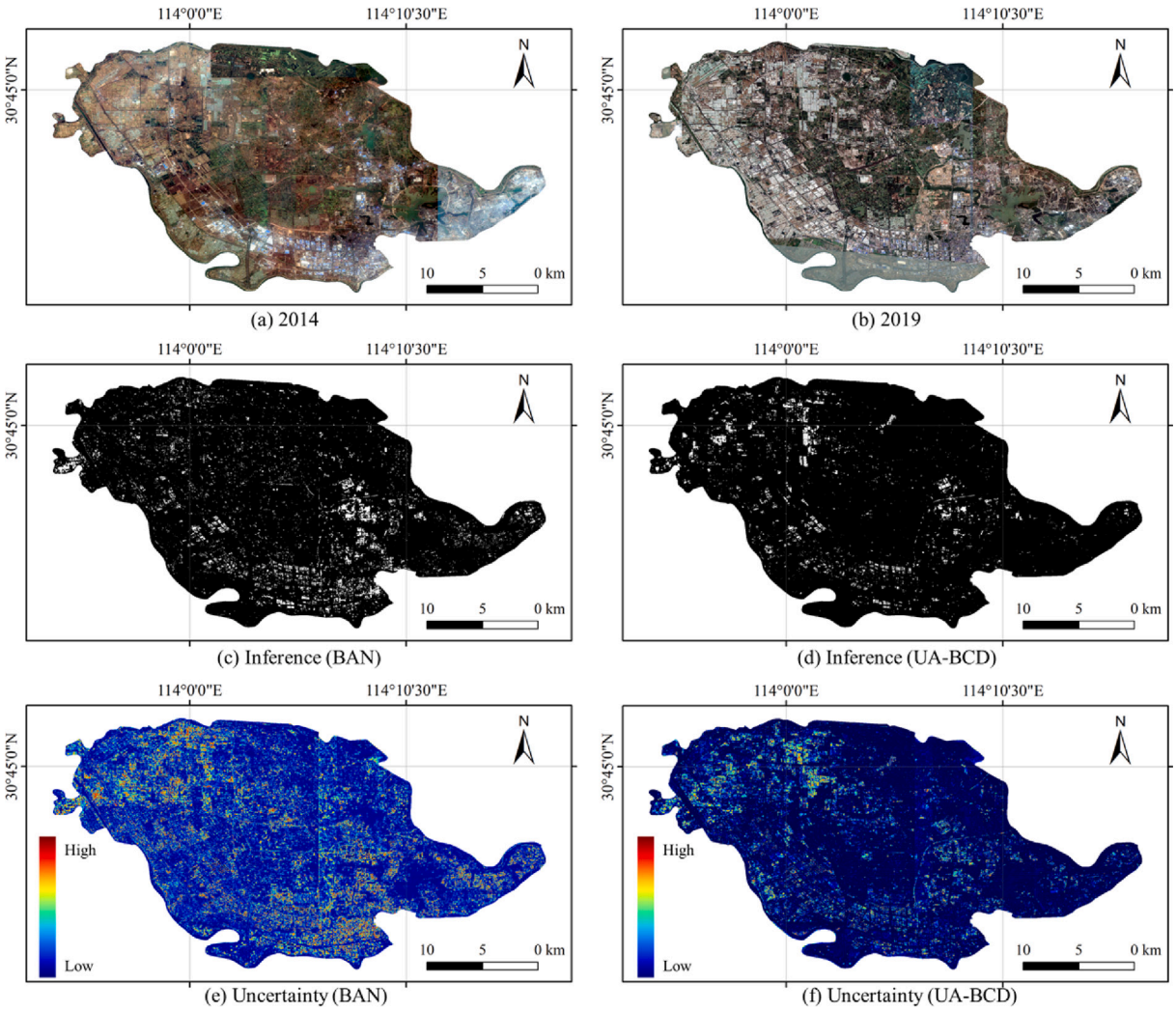


Fig. 16. Comparison of large-scale uncertainty maps from 2014 to 2019 generated by BAN (Li et al., 2024b) and our proposed UA-BCD framework.

of 1 meter (zoom level-18). The images from 2014 and 2019 were both captured on December 31 of their respective years. To fairly compare our method with existing BCD techniques, we used the Google-CD dataset (Peng et al., 2020) as the training set for mapping building changes in Dongxihu District.

For large-scale image inference, we employed a sliding window approach with multi-batch prediction. The input large image was first cropped into patches of size 256 × 256 pixels with a stride of 128 pixels. We then set a batch size (64 in this study) and read the cropped patches in multiple batches. After prediction, the patches were restored to the original size of the large image. This method effectively utilizes memory and significantly reduces inference time, completing the inference for the entire Dongxihu District in just 15 min.

5.2. Visual analysis

We applied our inference strategy to map building changes in Dongxihu District, Wuhan, for periods of 2014–2019 and 2019–2024, with a 5-year interval. For comparison, we selected the SOTA method BAN (Li et al., 2024b) and visualized the inference results. Additionally, we visualized the entropy values representing the uncertainty of the inference results, as shown in Figs. 16 and 17.

Significant changes in the urban landscape of Dongxihu District are observable from 2014 to 2019 and further to 2024. The differences in the images highlight the challenges in building mapping. Regarding uncertainty, our method exhibits significantly lower uncertainty compared with BAN, indicating that our approach effectively addresses the uncertainty problem in BCD mapping better than BAN. Moreover, there is a considerable difference in the mapping results between the two methods. Overall, BAN detects more building changes than our proposed UA-BCD. To further assess the mapping results, we conducted a visual comparison of specific areas for the periods 2014–2019 and 2019–2024, as illustrated in Figs. 18 and 19.

In the first scene in Fig. 18, the 2019 image with the cloud interference is not as clear as the 2014 image. Nevertheless, the building changes are not significant. The BAN method incorrectly identifies many unchanged buildings as changed, whereas our method accurately detects the correct building changes with significant superiority. In the second scene in Fig. 18, the red-boxed area represents a densely populated residential area that had not changed much from 2014 to 2019. BAN performs poorly here, whereas our UA-BCD precisely captures the change in only one building.

In Fig. 19, the first scene shows that BAN not only fails to detect the actual changes but also produces highly fragmented and incomplete results. By contrast, our method shows a clear advantage. In the second scene, BAN fails to overcome pseudo-changes caused by differences in building styles and image quality, leading to numerous misclassifications. Our method, on the contrary, effectively addresses these pseudo challenges.

This analysis evidences that our method outperforms BAN in terms of segmentation quality, reduction of false positives and false negatives, and lower uncertainty, which demonstrates the effectiveness

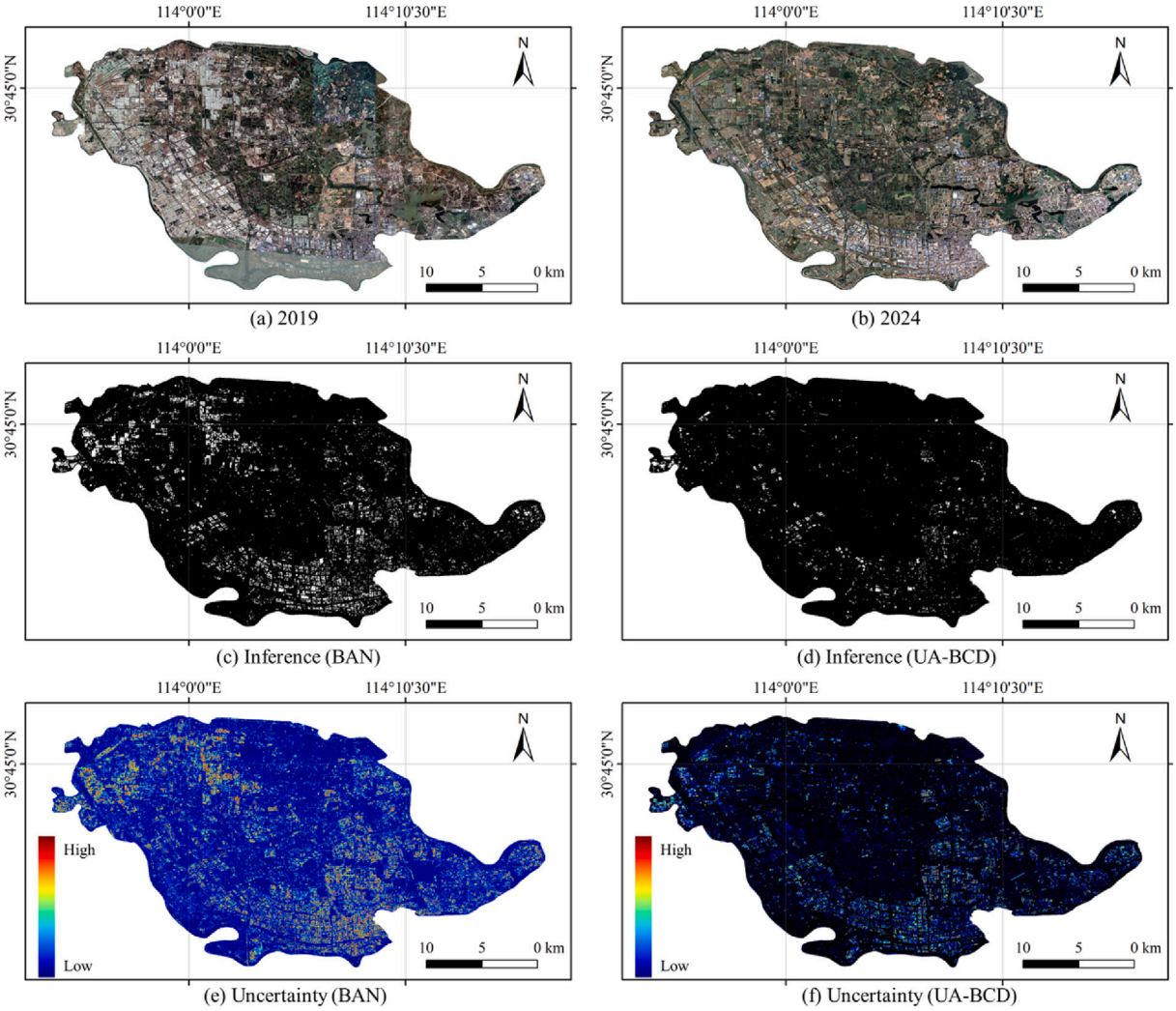


Fig. 17. Comparison of large-scale uncertainty maps from 2019 to 2024 generated by BAN (Li et al., 2024b) and our proposed UA-BCD framework.

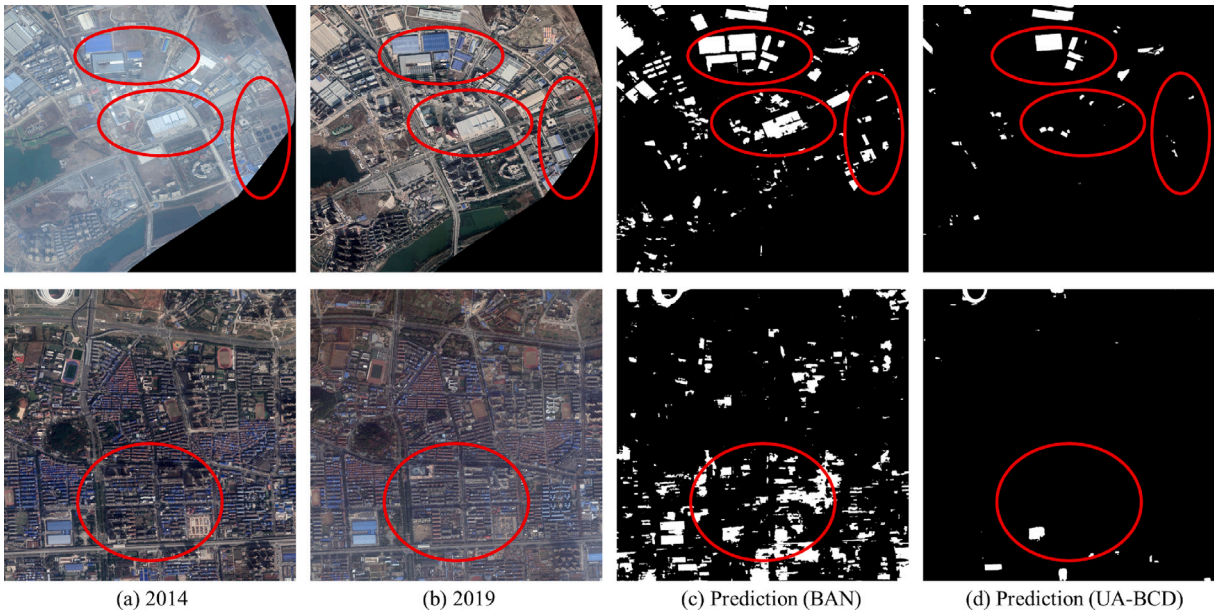


Fig. 18. Detailed comparison of large-scale BCD mapping results from 2014 to 2019 by BAN (Li et al., 2024b) and our proposed UA-BCD framework.

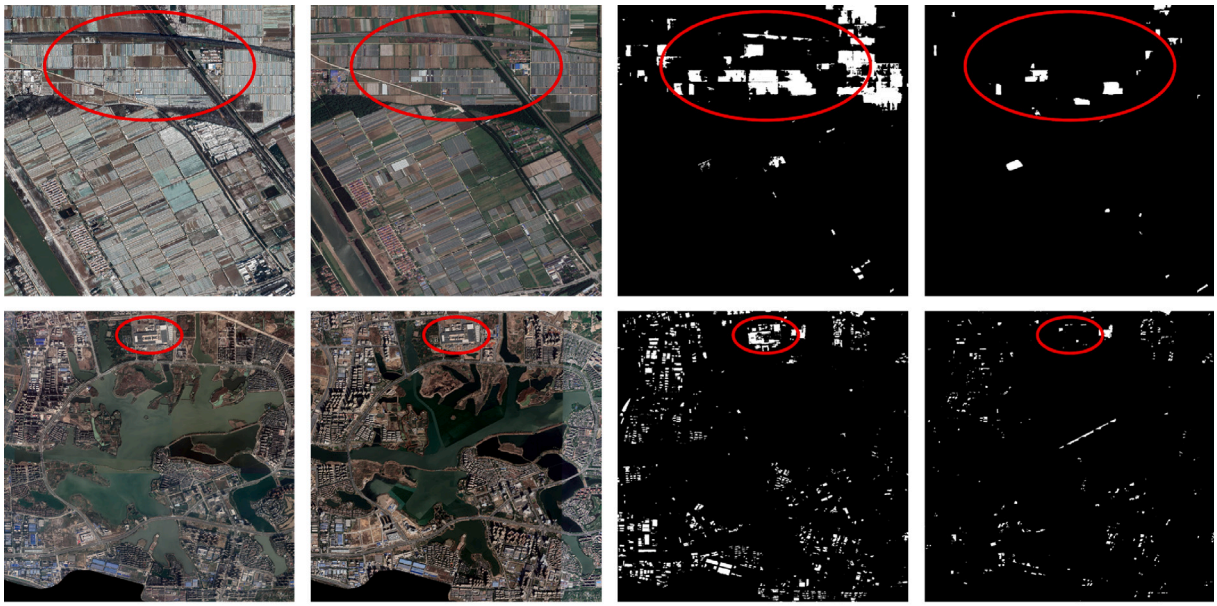


Fig. 19. Detailed comparison of large-scale BCD mapping results from 2019 to 2024 by BAN (Li et al., 2024b) and our proposed UA-BCD framework.

Table 9
Performance comparison with baseline models on the randomly sampled data in Dongxihu District.

Baseline	2014–2019 (%)			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
BAN	43.56	57.11	53.30	61.50
UA-BCD	**47.87**	**58.86**	**53.32**	**65.69**
Baseline	2019–2024 (%)			
	IoU ↑	*F1* ↑	*Pre* ↑	*Recall* ↑
BAN	47.05	63.99	72.06	57.55
UA-BCD	**54.08**	**68.11**	**77.05**	**61.03**

of addressing data-level aleatoric uncertainty and model-level epistemic uncertainty in our study, providing a highly reliable solution for real-world applications.

5.3. Quantitative evaluation analysis

Considering the lack of labeled data for evaluating the CD results in Dongxihu District during actual inference, we employed a random sampling method to select sample points for manual annotation and quantitative assessment of the inference results. Specifically, we divided the entire Dongxihu District image into non-overlapping patches of size 256 × 256 pixels. After excluding irrelevant boundary areas, we randomly selected 5% of the patches for annotation. The annotations included building changes from 2014 to 2019 and from 2019 to 2024. We then performed an accuracy assessment of the results.

As shown in Table 9, our method exhibits significant advantages over BAN in terms of IoU, F1, Precision, and Recall. This finding clearly indicates the superiority of our approach in large-scale applications, underscoring the critical importance of addressing uncertainty in BCD tasks.

6. Conclusion

The proposed UA-BCD framework has demonstrated its effectiveness in addressing the challenges of aleatoric and epistemic uncertainties in RS tasks. By incorporating a Siamese Pyramid Vision Transformer for feature extraction, an Aleatoric Uncertainty Estimation Module (AUEM) for quantifying and mitigating aleatoric uncertainty, a Knowledge-Guided Feature Enhancement Module (KGFEM) for refining features on the basis of a coarse CD map, and an Epistemic Uncertainty Estimator (EUE) to assess epistemic uncertainty, our UA-BCD framework can produce CD maps with lower uncertainty and higher confidence.

Experimental results on five large-scale public datasets have shown that UA-BCD outperforms existing CD methods, demonstrating its ability to handle complex contextual information and uneven sample distribution in RS images. The integration of uncertainty estimation and mitigation mechanisms into the CD process not only improves the accuracy and reliability of the results but also provides valuable insights into the underlying uncertainties of the task.

Additionally, we explored the potential of our proposed method for BCD in large urban areas under real-world scenarios. This application demonstrated significant advantages over current SOTA methods and provided an efficient solution for BCD mapping.

Future work can further explore the potential of UA-BCD in other RS applications, such as land cover classification and object detection. Additionally, the integration of more advanced uncertainty estimation techniques and the development of more robust decoding strategies could further enhance the performance of UA-BCD.

CRediT authorship contribution statement

Jiepan Li: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation. **Wei He:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Zhuohong Li:** Writing – review & editing, Visualization, Investigation. **Yujun Guo:** Writing – review & editing, Visualization. **Hongyan Zhang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Alcolea, A., Resano, J., 2022. Bayesian neural networks to analyze hyperspectral datasets using uncertainty metrics. IEEE Trans. Geosci. Remote Sens. 60, 1–10. http://dx.doi.org/10.1109/TGRS.2022.3205119.

Bandara, W.G.C., Patel, V.M., 2022. A transformer-based Siamese network for change detection. In: IGARSS 2022-2022 IEEE International Geoscience and Remote Sensing Symposium. IEEE, pp. 207–210.

Baumgartner, C.F., Tezcan, K.C., Chaitanya, K., Hötker, A.M., Muehlematter, U.J., Schawkat, K., Becker, A.S., Donati, O., Konukoglu, E., 2019. Phiseg: Capturing uncertainty in medical image segmentation. In: Medical Image Computing and Computer Assisted Intervention–MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13–17, 2019, Proceedings, Part II 22. Springer, pp. 119–127.

Chen, J., Gong, P., He, C., Pu, R., Shi, P., 2003. Land-use/land-cover change detection using improved change-vector analysis. Photogramm. Eng. Remote Sens. 69 (4), 369–379.

Chen, T., Lu, Z., Yang, Y., Zhang, Y., Du, B., Plaza, A., 2022a. A Siamese network based U-net for change detection in high resolution remote sensing images. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 15, 2357–2369.

Chen, H., Qi, Z., Shi, Z., 2021. Remote sensing image change detection with transformers. IEEE Trans. Geosci. Remote Sens. 60, 1–14.

Chen, H., Shi, Z., 2020a. A spatial-temporal attention-based method and a new dataset for remote sensing image change detection. Remote Sens. 12 (10), http://dx.doi.org/10.3390/rs12101662, URL https://www.mdpi.com/2072-4292/12/10/1662.

Chen, H., Shi, Z., 2020b. A spatial-temporal attention-based method and a new dataset for remote sensing image change detection. Remote Sens. 12 (10), 1662.

Chen, Z., Yang, B., Zhu, R., Dong, Z., 2024. City-scale solar PV potential estimation on 3D buildings using multi-source RS data: A case study in wuhan, China. Appl. Energy 359, 122720. http://dx.doi.org/10.1016/j.apenergy.2024.122720, URL https://www.sciencedirect.com/science/article/pii/S030626192400103X.

Chen, P., Zhang, B., Hong, D., Chen, Z., Yang, X., Li, B., 2022b. FCCDN: Feature constraint network for VHR image change detection. ISPRS J. Photogramm. Remote Sens. 187, 101–119.

Cheng, M., He, W., Li, Z., Yang, G., Zhang, H., 2024. Harmony in diversity: Content cleansing change detection framework for very-high-resolution remote-sensing images. ISPRS J. Photogramm. Remote Sens. 218, 1–19. http://dx.doi.org/10.1016/j.isprsjprs.2024.09.002, URL https://www.sciencedirect.com/science/article/pii/S092427162400340X.

Daudt, R.C., Le Saux, B., Boulch, A., 2018. Fully convolutional Siamese networks for change detection. In: 2018 25th IEEE International Conference on Image Processing. ICIP, IEEE, pp. 4063–4067.

Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., Fei-Fei, L., 2009. Imagenet: A large-scale hierarchical image database. In: 2009 IEEE Conference on Computer Vision and Pattern Recognition. IEEE, pp. 248–255.

Ding, L., Guo, H., Liu, S., Mou, L., Zhang, J., Bruzzone, L., 2022. Bi-temporal semantic reasoning for the semantic change detection in HR remote sensing images. IEEE Trans. Geosci. Remote Sens. 60, 1–14.

Fang, Y., Zhang, H., Yan, J., Jiang, W., Liu, Y., 2023. UDNet: Uncertainty-aware deep network for salient object detection. Pattern Recognit. 134, 109099.

Fung, T., LeDrew, E., 1987. Application of principal components analysis to change detection. Photogr. Eng. Remote Sens. 53 (12), 1649–1658.

Gao, S.-H., Cheng, M.-M., Zhao, K., Zhang, X.-Y., Yang, M.-H., Torr, P., 2019. Res2net: A new multi-scale backbone architecture. IEEE Trans. Pattern Anal. Mach. Intell. 43 (2), 652–662.

Gupta, R., Hosfelt, R., Sajeev, S., Patel, N., Goodman, B., Doshi, J., Heim, E., Choset, H., Gaston, M., 2019. Xbd: A dataset for assessing building damage from satellite imagery. arXiv preprint arXiv:1911.09296.

Han, C., Wu, C., Guo, H., Hu, M., Chen, H., 2023a. HANet: A hierarchical attention network for change detection with bi-temporal very-high-resolution remote sensing images. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 1–17. http://dx.doi.org/10.1109/JSTARS.2023.3264802.

Han, C., Wu, C., Guo, H., Hu, M., Li, J., Chen, H., 2023b. Change guiding network: Incorporating change prior to guide change detection in remote sensing imagery. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 16, 8395–8407. http://dx.doi.org/10.1109/JSTARS.2023.3310208.

Han, T., Wulder, M.A., White, J.C., Coops, N.C., Alvarez, M., Butson, C., 2007. An efficient protocol to process landsat images for change detection with tasselled cap transformation. IEEE Geosci. Remote Sens. Lett. 4 (1), 147–151.

Hasanlou, M., Seydi, S.T., 2018. Hyperspectral change detection: An experimental comparative study. Int. J. Remote Sens. 39 (20), 7029–7083.

He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 770–778.

Huang, H., He, W., Zhang, H., Xia, Y., Zhang, L., 2024. STFDiff: Remote sensing image spatiotemporal fusion with diffusion models. Inf. Fusion 102505.

Im, J., Jensen, J.R., 2005. A change detection model based on neighborhood correlation image analysis and decision tree classification. Remote Sens. Environ. 99 (3), 326–340.

Ji, S., Wei, S., Lu, M., 2018. Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set. IEEE Trans. Geosci. Remote Sens. 57 (1), 574–586.

Jiang, H., Hu, X., Li, K., Zhang, J., Gong, J., Zhang, M., 2020. PGA-SiamNet: Pyramid feature-based attention-guided Siamese network for remote sensing orthoimagery building change detection. Remote Sens. 12 (3), 484.

Krizhevsky, A., Sutskever, I., Hinton, G.E., 2012. Imagenet classification with deep convolutional neural networks. Adv. Neural Inf. Process. Syst. 25.

Li, K., Cao, X., Meng, D., 2024b. A new learning paradigm for foundation model-based remote sensing change detection. IEEE Trans. Geosci. Remote Sens. http://dx.doi.org/10.1109/TGRS.2024.3365825, 1–1.

Li, J., He, W., Cao, W., Zhang, L., Zhang, H., 2024a. UANet: an uncertainty-aware network for building extraction from remote sensing images. IEEE Trans. Geosci. Remote Sens. http://dx.doi.org/10.1109/TGRS.2024.3361211, 1–1.

Li, Z., He, W., Cheng, M., Hu, J., Yang, G., Zhang, H., 2023. Sinolc-1: the first 1 m resolution national-scale land-cover map of China created with a deep learning framework and open-access data. Earth Syst. Sci. Data 15 (11), 4749–4780. http://dx.doi.org/10.5194/essd-15-4749-2023.

Li, Z., He, W., Li, J., Lu, F., Zhang, H., 2024c. Learning without exact guidance: Updating large-scale high-resolution land cover maps from low-resolution historical labels. arXiv preprint arXiv:2403.02746.

Li, Z., Lu, F., Zhang, H., Tu, L., Li, J., Huang, X., Robinson, C., Malkin, N., Jojic, N., Ghamisi, P., Hänsch, R., Yokoya, N., 2022a. The outcome of the 2021 IEEE GRSS Data Fusion Contest—Track MSD: Multitemporal semantic change detection. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 15, 1643–1655. http://dx.doi.org/10.1109/JSTARS.2022.3144318.

Li, Z., Zhang, H., Lu, F., Xue, R., Yang, G., Zhang, L., 2022b. Breaking the resolution barrier: A low-to-high network for large-scale high-resolution land-cover mapping using low-resolution labels. ISPRS J. Photogramm. Remote Sens. 192, 244–267.

Lin, T.-Y., Dollár, P., Girshick, R., He, K., Hariharan, B., Belongie, S., 2017. Feature pyramid networks for object detection. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 2117–2125.

Lin, M., Yang, G., Zhang, H., 2022. Transition is a process: Pair-to-video change detection networks for very high resolution remote sensing images. IEEE Trans. Image Process. 32, 57–71.

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B., 2021. Swin transformer: Hierarchical vision transformer using shifted windows. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 10012–10022.

Liu, W., Lin, Y., Liu, W., Yu, Y., Li, J., 2023. An attention-based multiscale transformer network for remote sensing image change detection. ISPRS J. Photogramm. Remote Sens. 202, 599–609.

Long, J., Shelhamer, E., Darrell, T., 2015. Fully convolutional networks for semantic segmentation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 3431–3440.

Lu, D., Mausel, P., Brondizio, E., Moran, E., 2004. Change detection techniques. Int. J. Remote Sens. 25 (12), 2365–2401.

Ludeke, A.K., Maggio, R.C., Reid, L.M., 1990. An analysis of anthropogenic deforestation using logistic regression and GIS. J. Environ. Manag. 31 (3), 247–259.

Nielsen, A.A., Canty, M.J., 2008. Kernel principal component analysis for change detection. In: Image and Signal Processing for Remote Sensing XIV. Vol. 7109, SPIE, pp. 244–253.

Nielsen, A.A., Conradsen, K., Simpson, J.J., 1998. Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies. Remote Sens. Environ. 64 (1), 1–19.

Pan, J., Bai, Y., Shu, Q., Zhang, Z., Hu, J., Wang, M., 2024. M-Swin: Transformer-based multiscale feature fusion change detection network within cropland for remote sensing images. IEEE Trans. Geosci. Remote Sens. 62, 1–16. http://dx.doi.org/10.1109/TGRS.2024.3374421.

Peng, D., Bruzzone, L., Zhang, Y., Guan, H., Ding, H., Huang, X., 2020. SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images. IEEE Trans. Geosci. Remote Sens. 59 (7), 5891–5906.

Peng, D., Zhang, Y., Guan, H., 2019. End-to-end change detection for high resolution satellite images using improved UNet++. Remote Sens. 11 (11), 1382.

Quarmby, N., Cushnie, J., 1989. Monitoring urban land cover changes at the urban fringe from SPOT HRV imagery in south-east England. Int. J. Remote Sens. 10 (6), 953–963.

Rignot, E.J., Van Zyl, J.J., 1993. Change detection techniques for ERS-1 SAR data. IEEE Trans. Geosci. Remote Sens. 31 (4), 896–906.

Rosa, R.A., Fernandes, D., Nogueira, J.B., Wimmer, C., 2015. Automatic change detection in multitemporal X-and P-band SAR images using gram-Schmidt process. In: 2015 IEEE International Geoscience and Remote Sensing Symposium. IGARSS, IEEE, pp. 2797–2800.

Schwert, B., Rogan, J., Giner, N., Ogneva-Himmelberger, Y., Blanchard, S., Woodcock, C., 2013. A comparison of support vector machines and manual change detection for land-cover map updating in Massachusetts, USA. Remote Sens. Lett. 4 (9), 882–890.

Shi, Q., Liu, M., Li, S., Liu, X., Wang, F., Zhang, L., 2021. A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection. IEEE Trans. Geosci. Remote Sens. 60, 1–16.

Shu, Q., Pan, J., Zhang, Z., Wang, M., 2022. DPCC-net: Dual-perspective change contextual network for change detection in high-resolution remote sensing images. Int. J. Appl. Earth Obs. Geoinf. 112, 102940.

Simonyan, K., Zisserman, A., 2014. Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556.

Suresh, M., Jain, K., 2013. Change detection and estimation of illegal mining using satellite images. In: Proceedings of 2nd International Conference of Innovation in Electronics and Communication Engineering. ICIECE-2013.

Vázquez-Jiménez, R., Romero-Calcerrada, R., Novillo, C.J., Ramos-Bernal, R.N., Arrogante-Funes, P., 2017. Applying the chi-square transformation and automatic secant thresholding to landsat imagery as unsupervised change detection methods. J. Appl. Remote Sens. 11 (1), 016016–016016.

Wang, Z., Peng, C., Zhang, Y., Wang, N., Luo, L., 2021b. Fully convolutional Siamese networks based change detection for optical aerial images with focal contrastive loss. Neurocomputing 457, 155–167.

Wang, W., Xie, E., Li, X., Fan, D.-P., Song, K., Liang, D., Lu, T., Luo, P., Shao, L., 2021a. Pyramid vision transformer: A versatile backbone for dense prediction without convolutions. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 568–578.

Wang, W., Xie, E., Li, X., Fan, D.-P., Song, K., Liang, D., Lu, T., Luo, P., Shao, L., 2022. Pvt v2: Improved baselines with pyramid vision transformer. Comput. Vis. Media 8 (3), 415–424.

Xie, S., Girshick, R., Dollár, P., Tu, Z., He, K., 2017. Aggregated residual transformations for deep neural networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 1492–1500.

Xu, Y., Wu, L., Liu, X., Chen, C., Li, Y., Zhang, Q., Liu, M., Zhang, T., Yang, B., 2024. Remote sensing change detection method based on dynamic adaptive focal loss. IEEE Trans. Geosci. Remote Sens..

Yang, F., Zhai, Q., Li, X., Huang, R., Luo, A., Cheng, H., Fan, D.-P., 2021. Uncertainty-guided transformer reasoning for camouflaged object detection. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 4146–4155.

Yin, H., Weng, L., Li, Y., Xia, M., Hu, K., Lin, H., Qian, M., 2023. Attention-guided Siamese networks for change detection in high resolution remote sensing images. Int. J. Appl. Earth Obs. Geoinf. 117, 103206.

Zhang, J., Fan, D.-P., Dai, Y., Anwar, S., Saleh, F.S., Zhang, T., Barnes, N., 2020b. UC-Net: Uncertainty inspired RGB-d saliency detection via conditional variational autoencoders. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition. CVPR, pp. 8579–8588. http://dx.doi.org/10.1109/CVPR42600.2020.00861.

Zhang, X., Xiao, P., Feng, X., Yuan, M., 2017. Separate segmentation of multi-temporal high-resolution remote sensing images for object-based change detection in urban area. Remote Sens. Environ. 201, 243–255.

Zhang, C., Yue, P., Tapete, D., Jiang, L., Shangguan, B., Huang, L., Liu, G., 2020a. A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images. ISPRS J. Photogramm. Remote Sens. 166, 183–200.

Zhang, Y., Zhang, J., Hamidouche, W., Deforges, O., 2023. Predictive uncertainty estimation for camouflaged object detection. IEEE Trans. Image Process. 32, 3580–3591.

Zheng, Z., Zhong, Y., Wang, J., Ma, A., Zhang, L., 2021. Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters. Remote Sens. Environ. 265, 112636.

Zhou, Z., Rahman Siddiquee, M.M., Tajbakhsh, N., Liang, J., 2018. Unet++: A nested u-net architecture for medical image segmentation. In: Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: 4th International Workshop, DLMIA 2018, and 8th International Workshop, ML-CDS 2018, Held in Conjunction with MICCAI 2018, Granada, Spain, September 20, 2018, Proceedings 4. Springer, pp. 3–11.